

Department of Electrical, Electronic and Computer Engineering

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Design and Implementation of a Secure Windows Server 2019 Infrastructure for Enterprise Network Services

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by

Group 1

Group project report submitted in partial fulfilment of the requirements for the degree

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in the Faculty of Engineering and the Built Environment

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Examiner: Mr. Vusumuzi Moyo
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Abstract

This report presents the installation, configuration, and testing of a Windows Server 2019 environment developed to meet the network and security requirements of CPUT. The project was implemented using the hypervisor utilised during Semester 1 and included the deployment of several key server roles and services.

The implementation involved configuring Active Directory Domain Services (AD DS) for centralized user and resource management, Domain Name System (DNS) for name resolution, and Dynamic Host Configuration Protocol (DHCP) for automatic IP address allocation. In addition, Group Policy settings were applied to improve the security of the environment by restricting Command Prompt access, disabling removable storage devices, preventing unauthorized software installations, and enforcing stronger password policies.

Remote Access Services and Windows Certificate Services were also configured to provide secure connectivity and authentication within the network environment. Each service was tested to verify that it functioned correctly and met the specified requirements. Screenshots of all installation, configuration, and testing procedures were captured and included as evidence of successful implementation.

The results demonstrate how Windows Server 2019 can be used to create a secure and manageable network infrastructure that supports organisational requirements while following recommended security practices.

Keywords: CPUT, Windows Server 2019, Active Directory, DNS, DHCP, Group Policy, Remote Access, Certificate Services, Network Administration, Security Policies, Server Configuration

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1. Introduction

As organisations continue to grow, the need for a reliable and secure network infrastructure becomes increasingly important. CPUT requires an IT environment that can support its expanding operations while ensuring efficient management of users, devices, and network resources. To meet these requirements, a Windows Server 2019 environment was implemented using the hypervisor used during Semester 1.

This project focuses on the installation, configuration, and testing of key Windows Server services, including Active Directory Domain Services (AD DS), DNS, DHCP, Group Policy, Remote Access, and Windows Certificate Services. Screenshots were captured throughout the process to provide evidence of the installation and configuration steps performed.

1.1 Background

Windows Server 2019 is widely used in organisations to manage network resources, user accounts, security policies, and shared services from a central location. Services such as Active Directory, DNS, and DHCP help administrators manage users and devices more efficiently while ensuring that network communication functions correctly.

Security is also an important part of network administration. For this reason, several security policies were implemented in this project. These include disabling Command Prompt access, preventing the use of removable storage devices, restricting software installations, enforcing a minimum password length of ten characters, and requiring users to change their passwords regularly. These measures help reduce security risks and improve the overall protection of the network environment.

1.2 Objectives

The objectives of this report are:

- To install and configure Windows Server 2019.
- To configure and test Active Directory Domain Services (AD DS).

- To configure and test DHCP services.
- To configure and test DNS services.
- To implement Group Policy settings that improve system security.
- To configure Remote Access services.
- To install and configure Windows Certificate Services.
- To verify that all configured services are functioning correctly.
- To document all installation, configuration, and testing procedures using screenshots.

1.3 Report Structure

The remainder of this report is organised as follows:

- Chapter 2 covers the installation and initial configuration of Windows Server 2019.
- Chapter 3 discusses the configuration and testing of Active Directory Domain Services (AD DS).
- Chapter 4 explains the configuration and testing of DHCP services.
- Chapter 5 presents the configuration and testing of DNS services.
- Chapter 6 describes the implementation of Group Policy settings.
- Chapter 7 discusses the configuration of Remote Access services.
- Chapter 8 explains the installation and configuration of Windows Certificate Services.
- Chapter 9 presents the testing and verification of the configured services.
- Chapter 10 provides the conclusion of the project.
- The final section contains the references used in this report.

2. Installation and Initial Configuration of Windows Server 2019

2.1 Introduction

This chapter describes the installation and initial configuration of Windows Server 2019 within the virtualised environment used during Semester 1. The purpose of this stage was to establish a functioning server that could support the deployment of the network services required for the project.

The installation process involved preparing the virtual machine using VirtualBox, installing the operating system, and completing the initial server configuration. These steps provided the foundation for the subsequent configuration of Active Directory Domain Services (AD DS), DNS, DHCP, Group Policy, Remote Access, and Windows Certificate Services.

2.2 Virtual Machine Preparation

Before installing Windows Server 2019, a virtual machine was prepared using VirtualBox. Virtualisation allows multiple operating systems to run on a single physical computer while maintaining separate virtual environments.

The virtual machine was configured with the necessary resources to support the operating system and the planned server roles. These resources included virtual memory, processor allocation, virtual storage, and a network adapter. The Windows Server 2019 installation media was then attached to the virtual machine, allowing it to boot from the installation image.

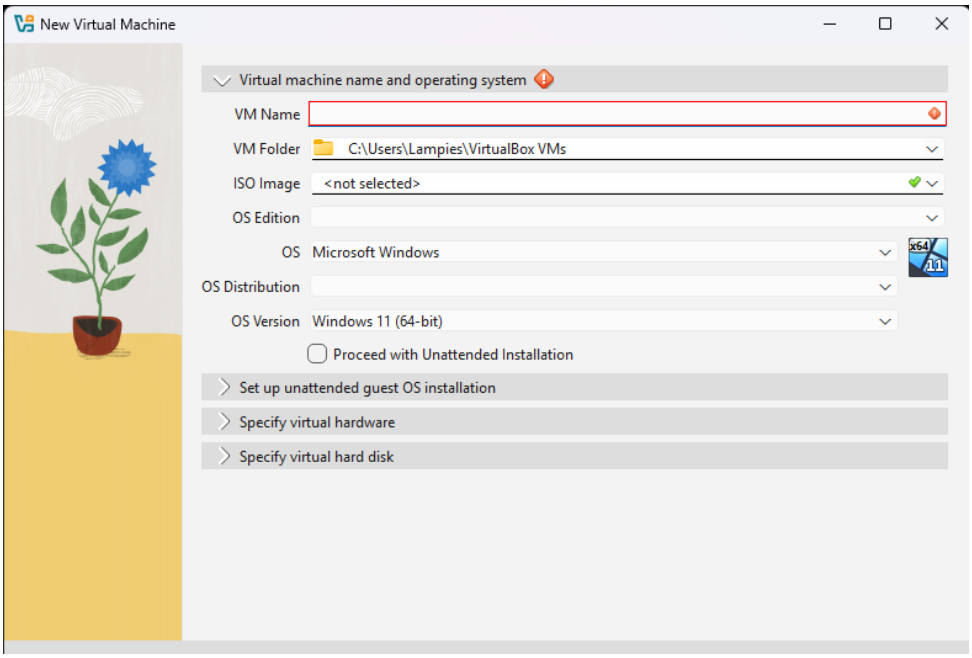


Figure 2.1: Initial Virtual Machine Configuration.

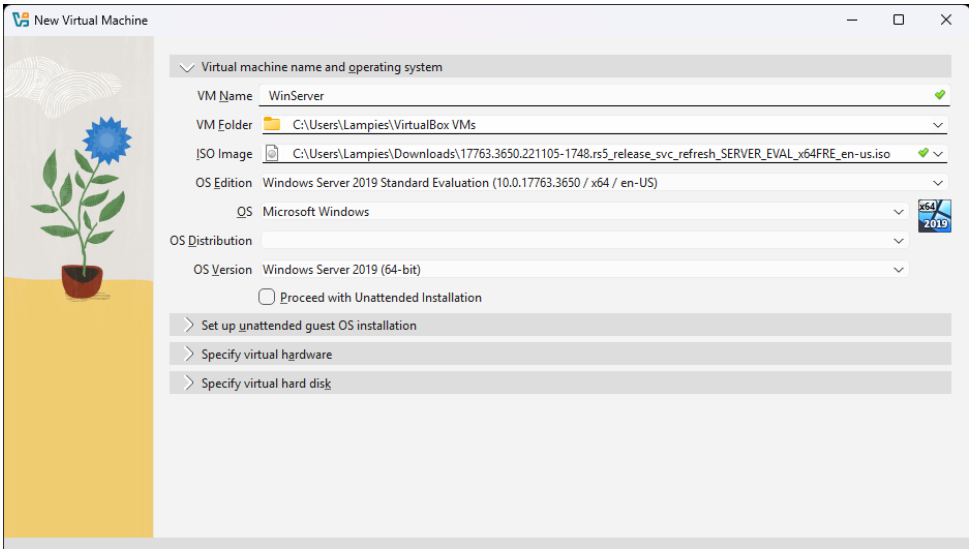


Figure 2.2: Virtual Machine Configuration Step 2.

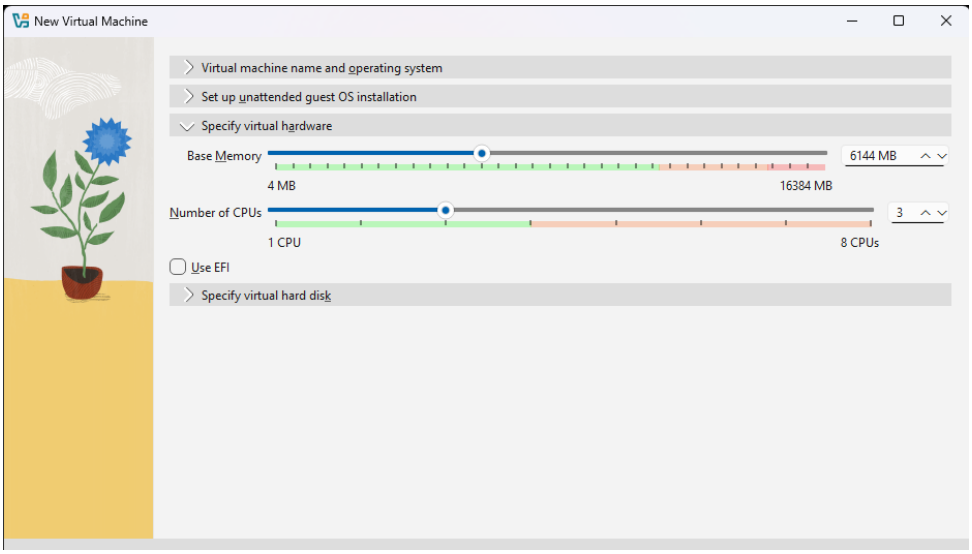


Figure 2.3: Virtual Machine Configuration Step 3.

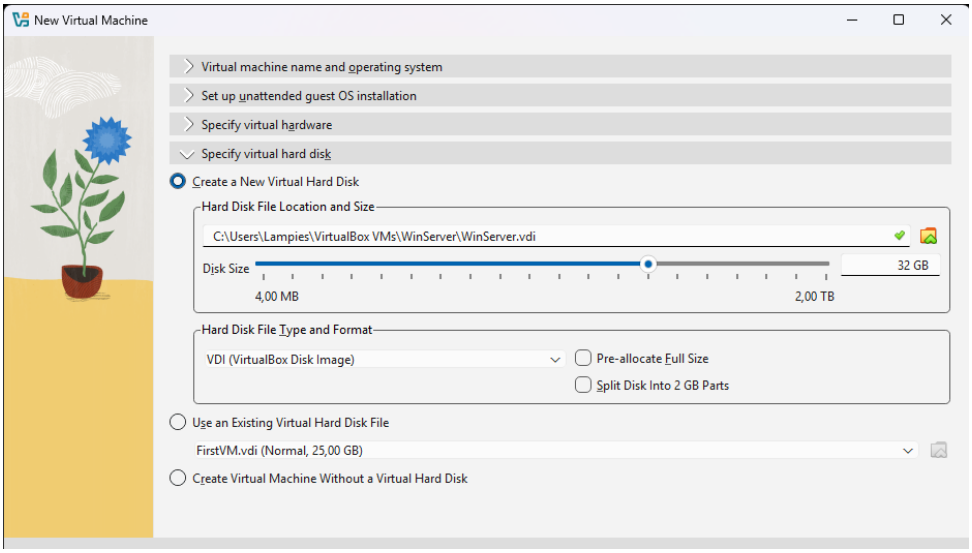


Figure 2.4: Virtual Machine Configuration Step 4.

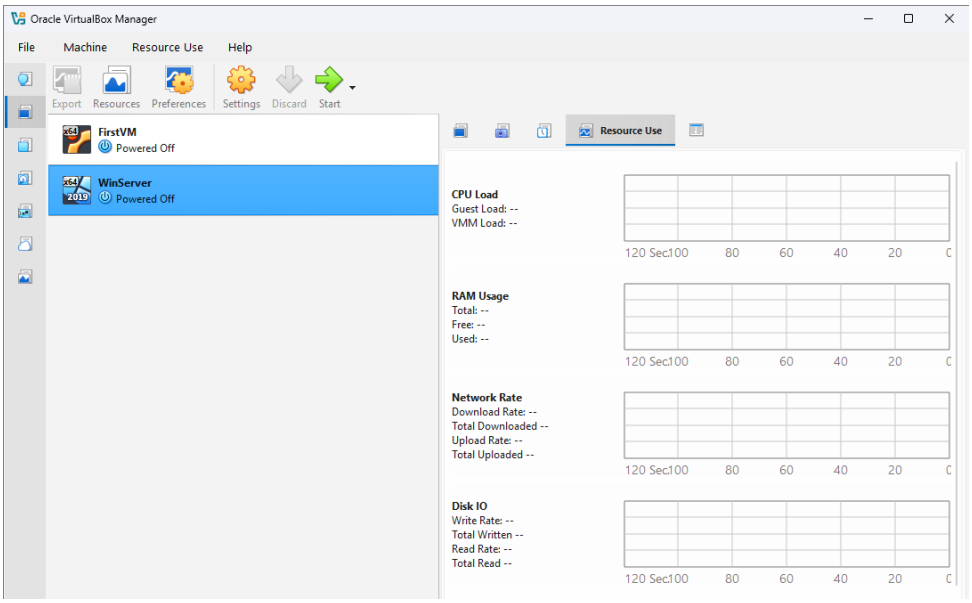


Figure 2.5: Virtual Machine Configuration Step 5.

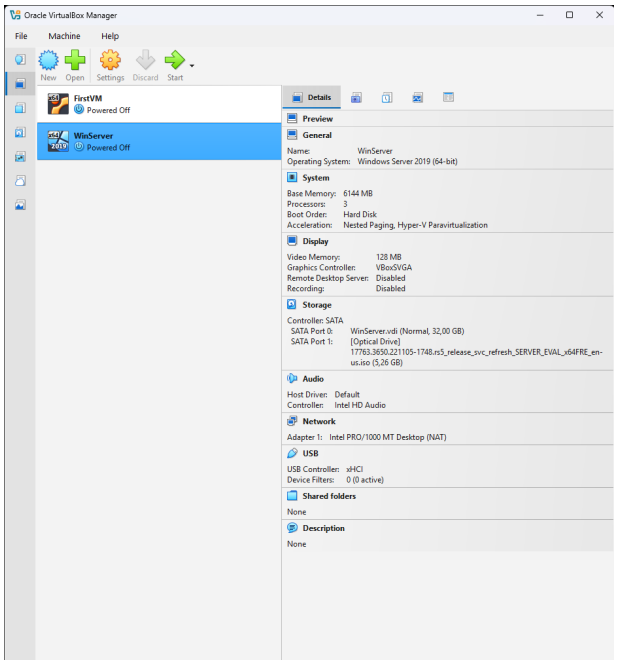


Figure 2.6: Final Virtual Machine Configuration Step.

2.3 Installation of Windows Server 2019

Once the virtual machine had been configured, it was started and the Windows Server 2019 installation process was initiated. The installation wizard was used to select the appropriate language, time and currency format, and keyboard layout.

The required Windows Server 2019 edition was then selected. The installation proceeded by accepting the applicable licence terms, selecting the installation type,

and choosing the virtual disk on which the operating system would be installed.

After the installation files had been copied and the required setup operations completed, the virtual machine restarted. The initial administrator password was configured, allowing access to the server using the local Administrator account.

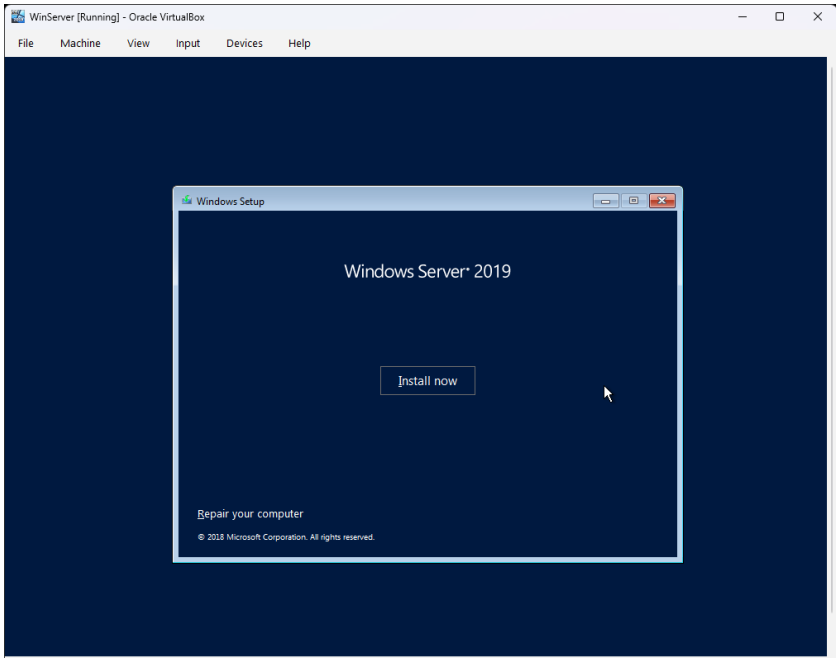


Figure 2.7: Installation Screen 1.

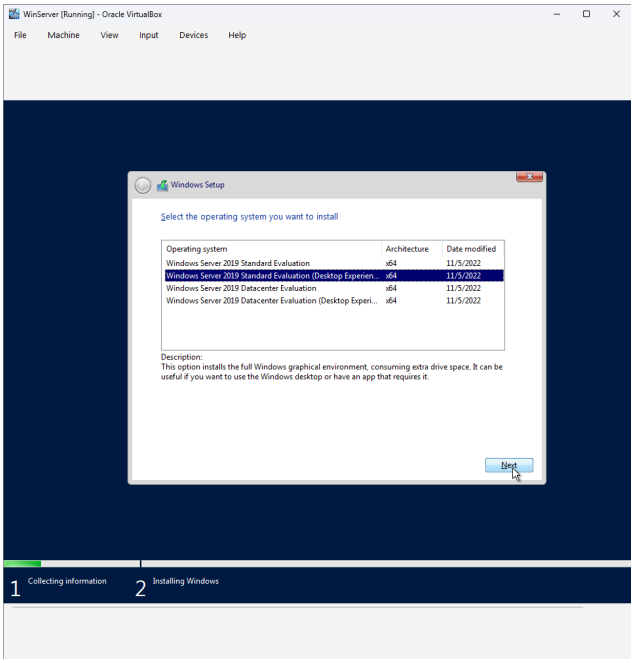


Figure 2.8: Installation Screen 2.

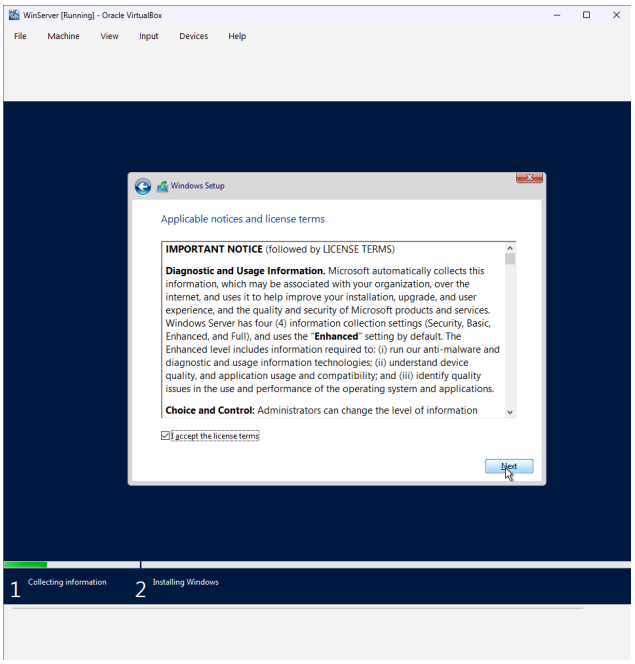


Figure 2.9: Installation Screen 3.

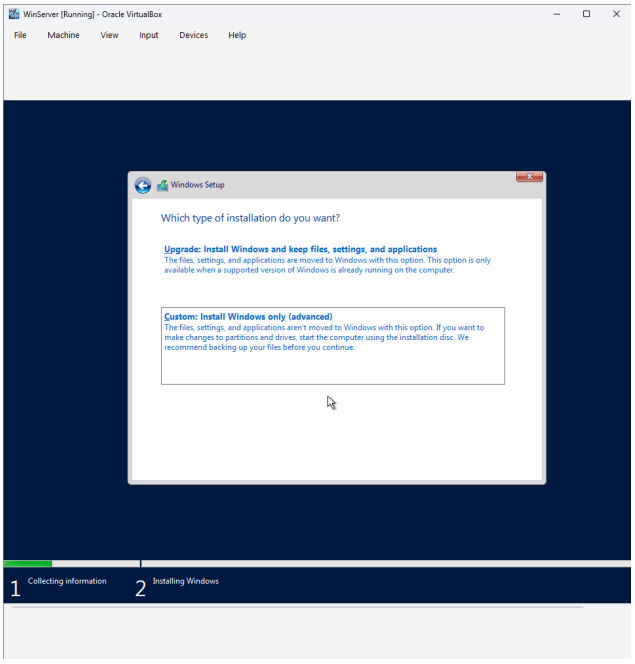


Figure 2.10: Installation Screen 4.

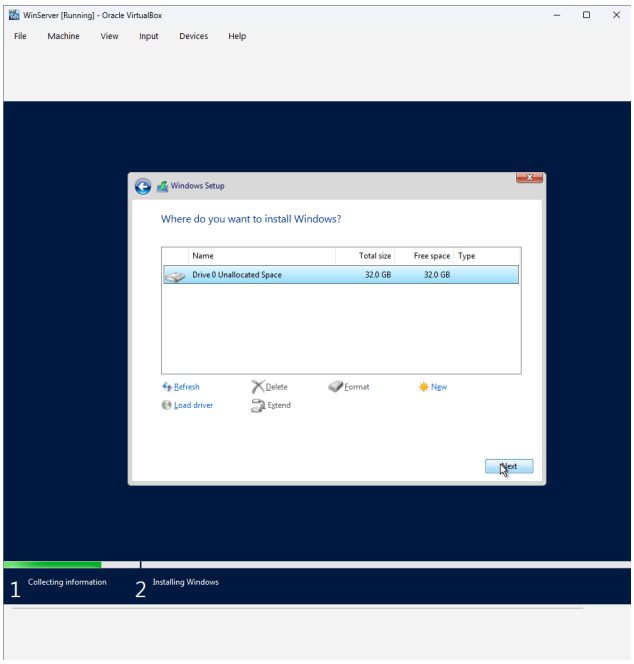


Figure 2.11: Installation Screen 5.

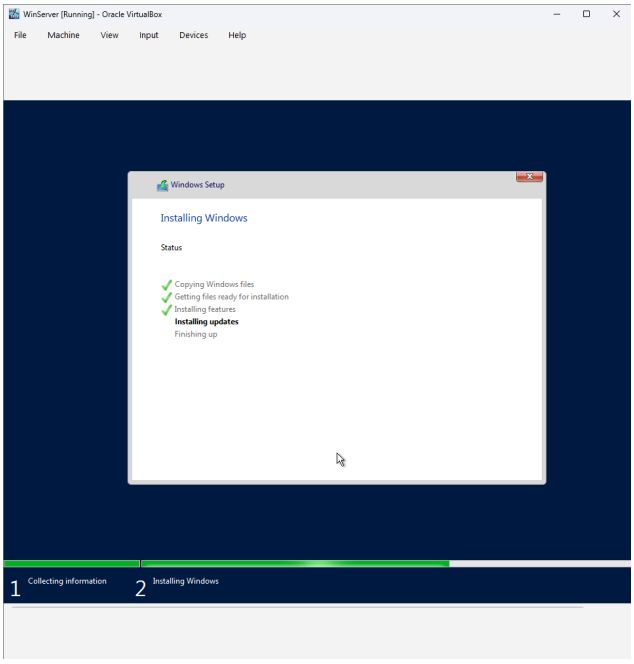


Figure 2.12: Installation Screen 6.

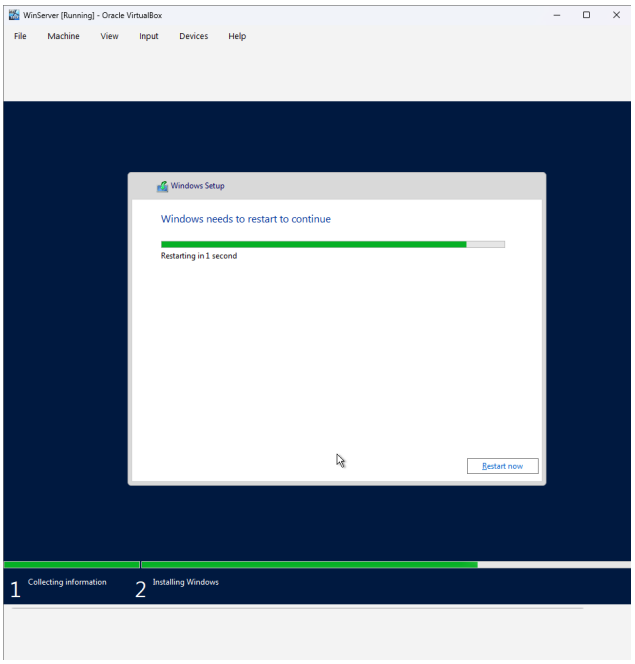


Figure 2.13: Installation Screen 7.

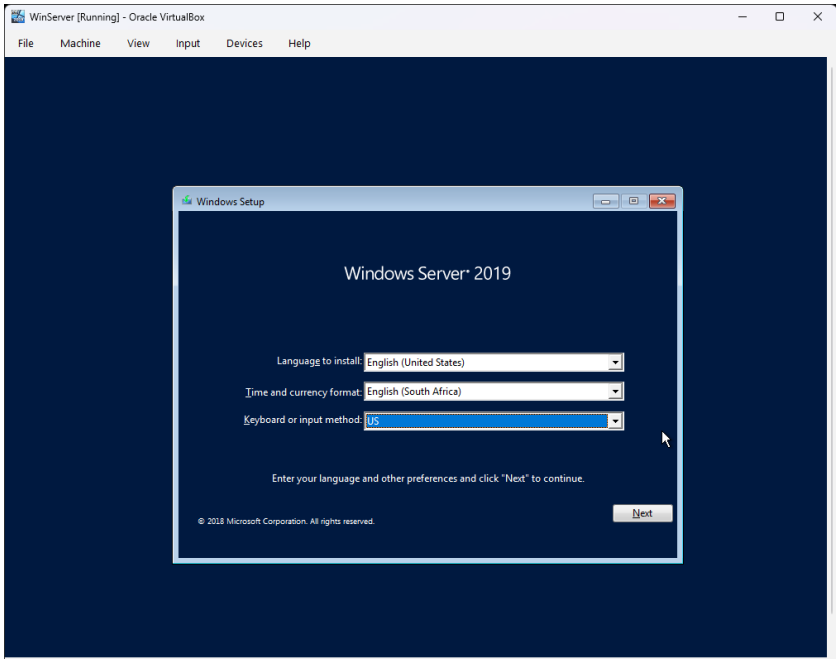


Figure 2.14: Installation Screen 8.

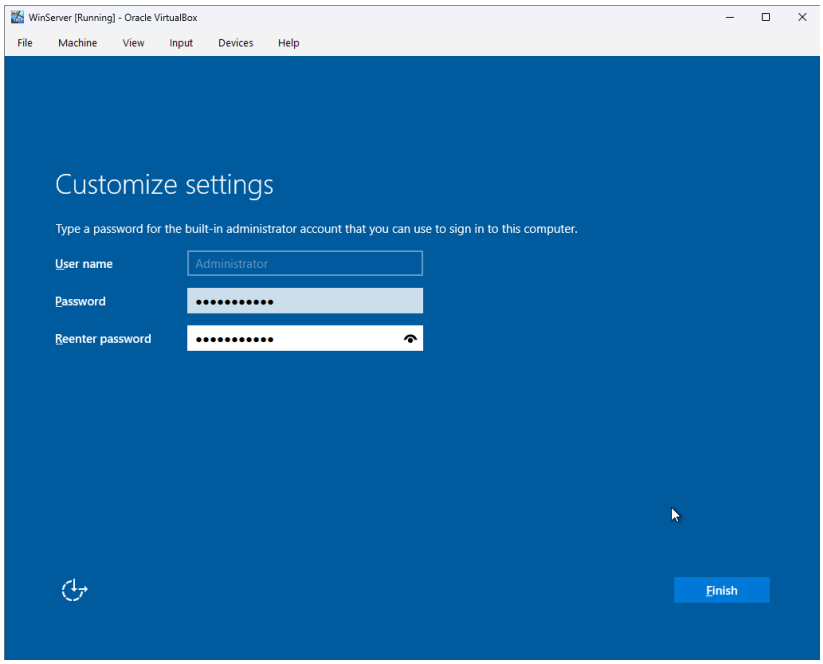


Figure 2.15: Windows Server Account Set-Up.

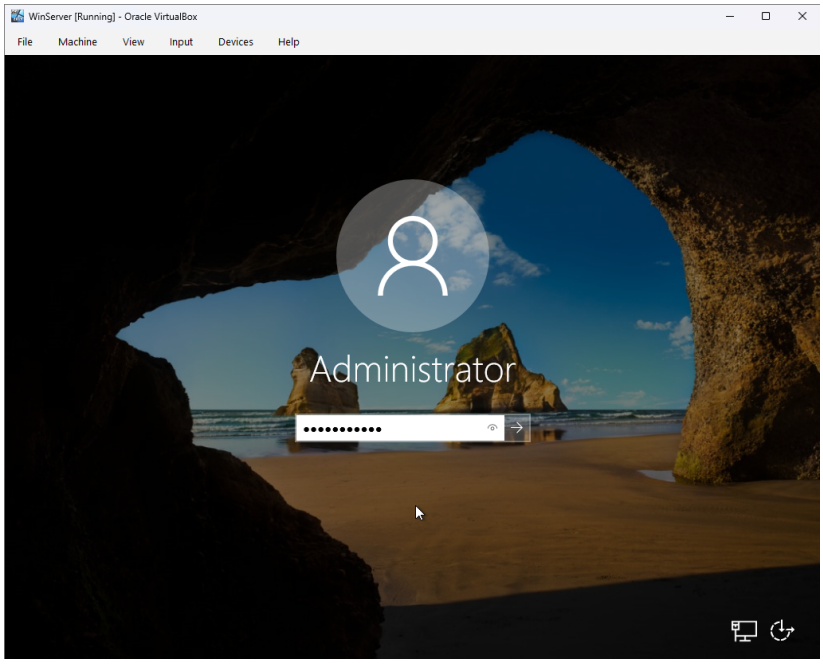


Figure 2.16: Windows Server Log-In Screen.

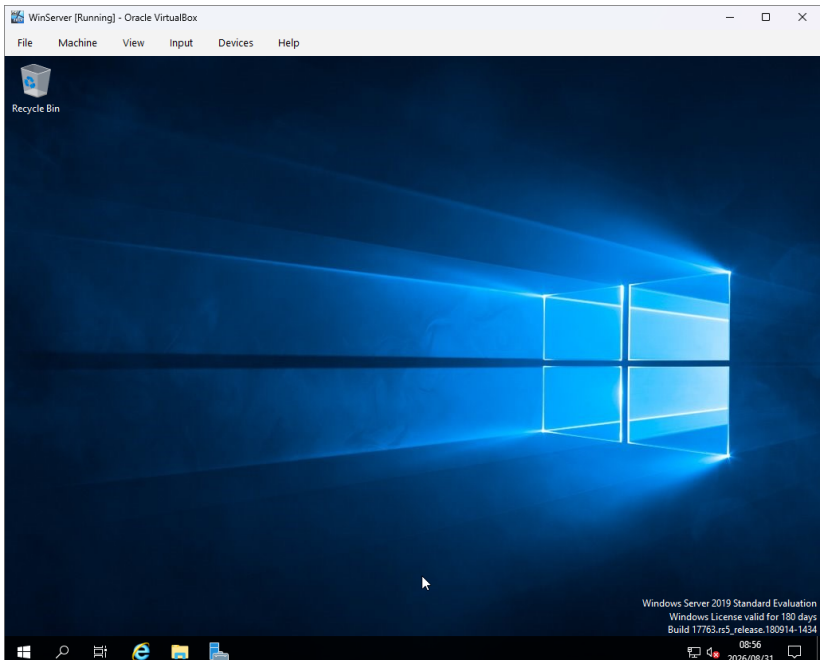


Figure 2.17: Windows Server Loaded and installed.

2.4 Testing

Testing was conducted to verify that Windows Server 2019 had been installed successfully and that the initial server configuration was functioning as expected. The tests focused on operating system availability, server identification, network configuration, and basic network connectivity.

2.4.1 Operating System Verification

The server was restarted to confirm that Windows Server 2019 could boot successfully after installation. The administrator was able to log in, and the Server Manager dashboard was accessible. This confirmed that the operating system was operational and ready for further configuration.

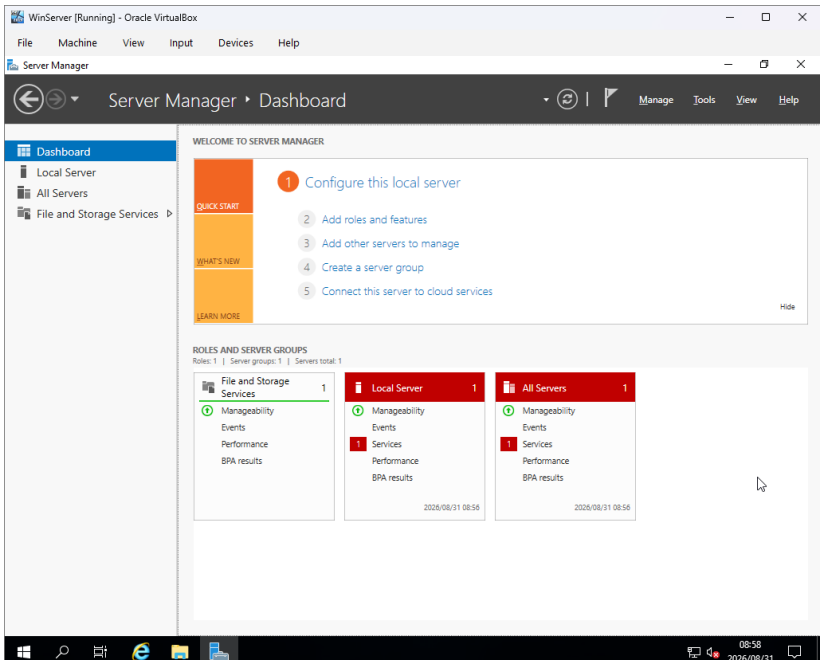


Figure 2.18: Server Manager dashboard following the installation of Windows Server 2019.

The dashboard provided access to the server's management tools and the interface used to install and configure additional server roles and features.

Further testing should verify the server hostname, IP configuration, and network connectivity. The following commands can be used to perform these checks:

- `hostname` – displays the configured computer name.
- `ipconfig /all` – displays the network adapter configuration.
- `ping <gateway-IP>` – tests connectivity to the configured network gateway, where applicable.

The outputs should be recorded and included as supporting evidence. Any failed tests should be investigated and resolved before proceeding with the configuration of additional server roles.

More comprehensive testing of Active Directory Domain Services, DNS, DHCP, Remote Access, and Windows Certificate Services is addressed in the relevant chapters of this report.

2.5 Chapter Summary

This chapter described the preparation of the virtual machine, installation of Windows Server 2019, and completion of the initial server configuration. The process established the operating system environment required for the deployment of additional server roles and network services.

Initial verification procedures were also described to assess operating system availability, server identification, network configuration, and connectivity. These steps establish the foundation for the remaining stages of the project.

The next chapter discusses the configuration and testing of Active Directory Domain Services (AD DS), which provides centralised management of users, computers, and authentication within the network environment.

3. DHCP Configuration

3.1 Background

Every machine, from computers to servers, requires an IP address for it to be able to send and receive packets of data or information. ([1]) These IP addresses act like home addresses on a street, and whilst DNS is responsible for resolving those IP addresses into names, there are two ways of assigning IP addresses to machines. These methods are static addressing and DHCP ([1]).

Static Addressing is the process of configuring IP addresses on a system manually. It is a hands-on approach where IP addresses are physically plugged into the NIC settings on a particular device ([1]). Despite its simplicity, this method is not scalable in environments that host multiple devices like a small or large company, where there are laptops, desktops, printers, servers or other types of machines ([1]). Each of these devices needs an IP address, however manually assign them becomes tedious at a larger scale. The solution to this problem lies in the Dynamic Host Configuration Protocol (DHCP) ([1]).

DHCP is a protocol that solves the problem of providing IP addresses for many devices on a wide scale. It allows for machines or devices that are plugged into the network and obtain IP address information ([1]). This ability to assign IP address information to devices allows for DHCP to have multiple applications. When a laptop or phone is connected to a Wi-Fi router to gain internet access, a DHCP server allows for that device to route traffic on the network by assigning IP addressing information ([1]). When a device connects to public Wi-Fi, its DHCP server runs on the router itself and in businesses where Windows Server serves as a datacenter, DHCP services are most often hosted on one or more servers across the network ([1]).

3.2 Configuration Steps

3.2.1 DHCP Post-Install configuration

Like other roles, the DHCP server needs to be provided a role during the installation phase. After the DHCP role has been added to the server, it isn't yet properly configured.

The DHCP server can give clients two kinds of access groups: DHCP Administrators and DHCP Users ([2]). DHCP Users have read-only access to the server, which allows them to view, but not to modify server data, including DHCP server configuration, registry keys, DHCP log files, and the database ([2]). They cannot create scopes, modify options values, create reservations or exclusion ranges, and more. DHCP Administrators are able to view and modify data at the DHCP server ([2]). They can create and delete scopes, add reservations, change option values, create superscopes or perform any other activity needed to adminster the DHCP server ([2]). Despite this overseering control, they do not have unlimited administrative rights,because they on have rights on the local computer only ([2]).

These two groups are local to the server as well, meaning they cannot authorize DHCP Servers. This can only be done by Domain Admins([2]). Authorizing the server is needed because DHCP communication typically occurs before any user or computer authentiation. DHCP protocols are based on IP protocols as well, therefore, an unknown DHCP server could provide invalid information to clients. Authorization prevents this ([3]).

The authorization of the DHCP server in the wizard required for users to be given credentials that they would need to access the server on a user level. So far the only credential set was `OPS\Administrator`. For any users who attempt to connect to the server in future, they must use this username.

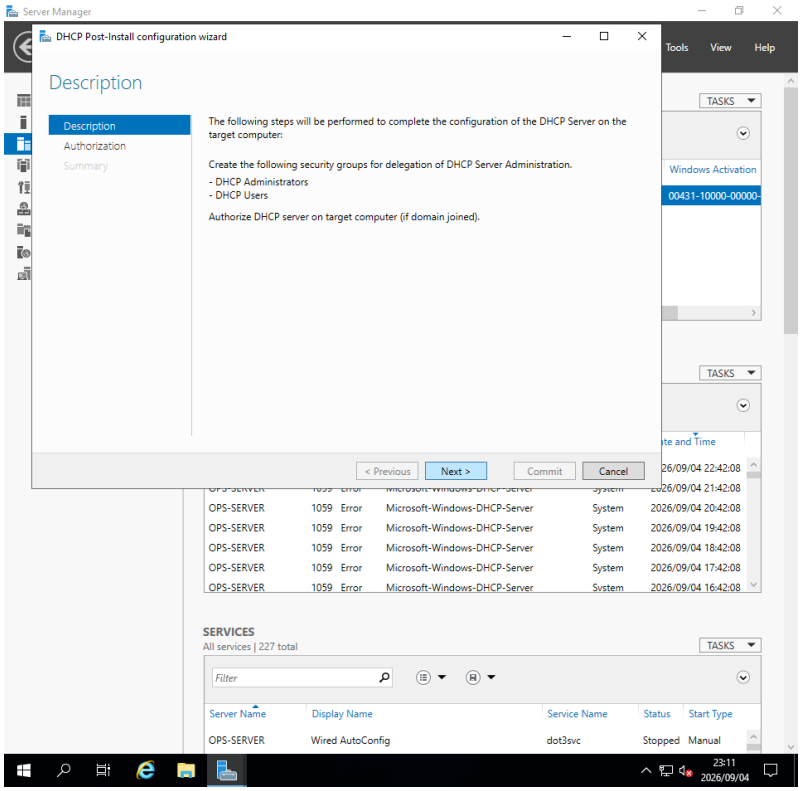


Figure 3.1: Description of DHCP Post-Install Configuration Wizard

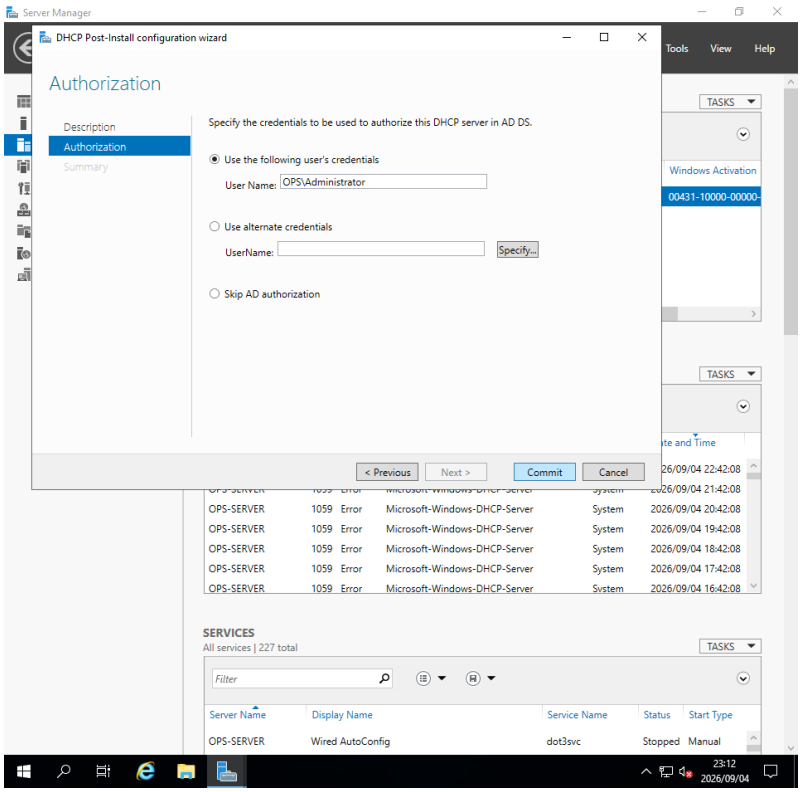


Figure 3.2: Authorization of DHCP Post-Install Configuration Wizard

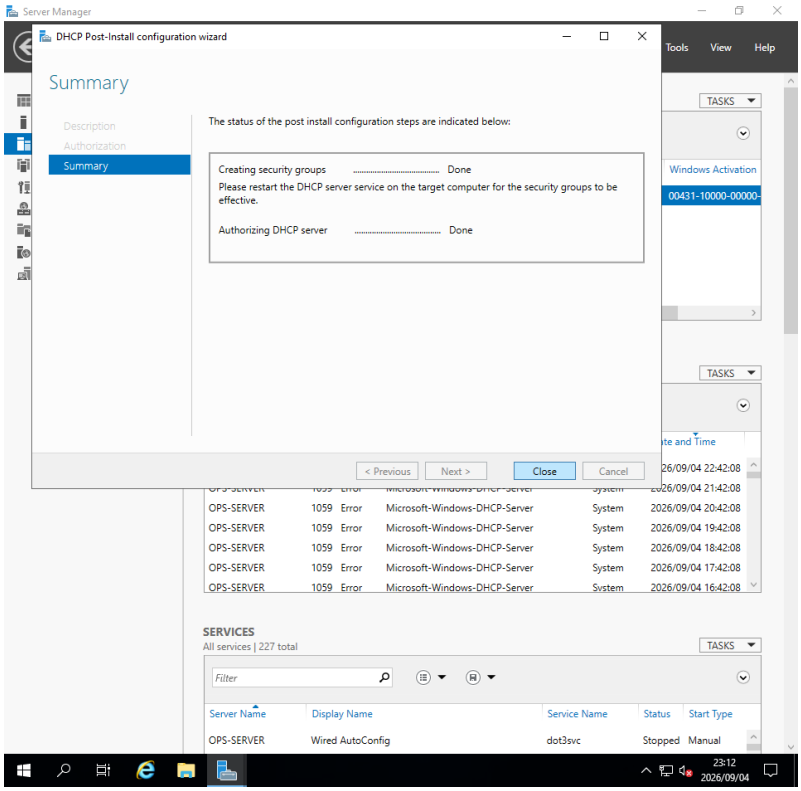


Figure 3.3: Post Install Configuration Completed

3.2.2 The DHCP Scope

For a DHCP server to hand out IP addresses, it needs to know IP addresses, subnet mask, default gateway and DNS server addresses are within the network so that it can package and hand out the information to computers who request it. This package of information is called a DHCP scope ([1]).

To begin defining the scope, we decided to launch a management tool using the Tools menu inside the Server Manager. This will open up a window for DHCP with the name of the device that the server is located on.

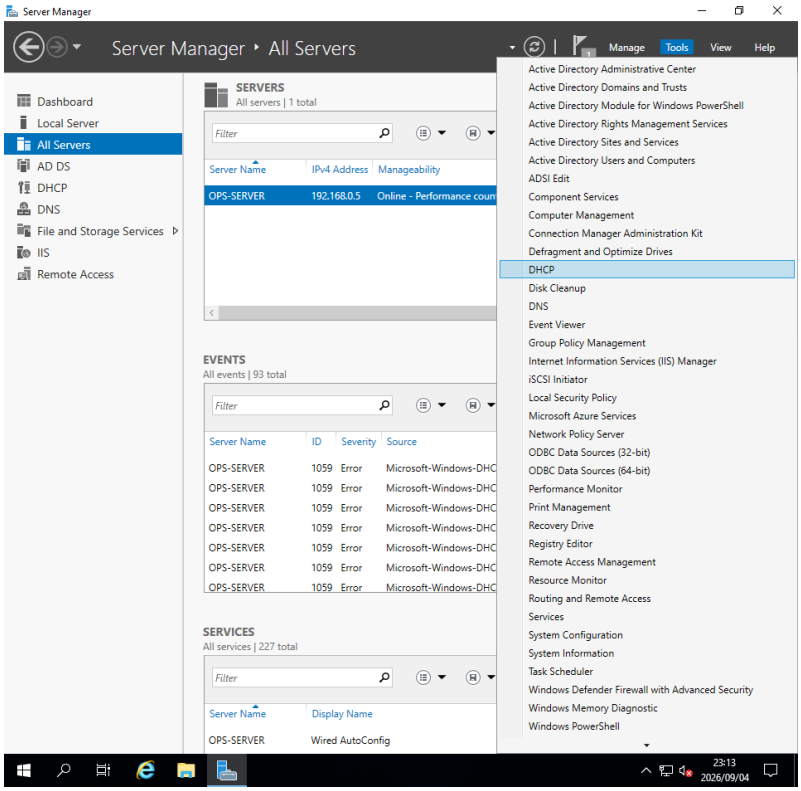


Figure 3.4: DHCP Selected on Tools Dropdown Menu

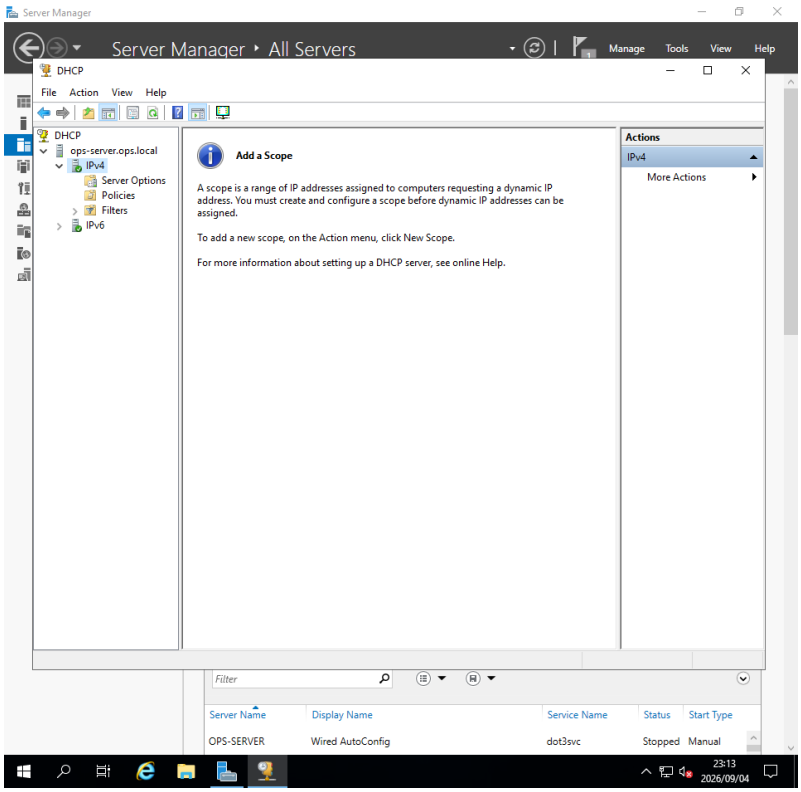


Figure 3.5: DHCP Windows

After opening the DHCP window, the computer was selected and dropped down two protocols: IPv4 and IPv6. The former was selected and first left-clicked on, where it gave information regarding the addition of a scope, along with the server options, policies and filters. This protocol did not have a scope as yet, so it needed to be configured. To properly begin the process, IPv4 was right-clicked on, giving a drop-down menu when the option to add a new scope was present. Choosing this option opened the New Scope Wizard. After clicking next on the wizard, the basics of the scope needed to be decided. The name given was GROUP 1-SCOPE.

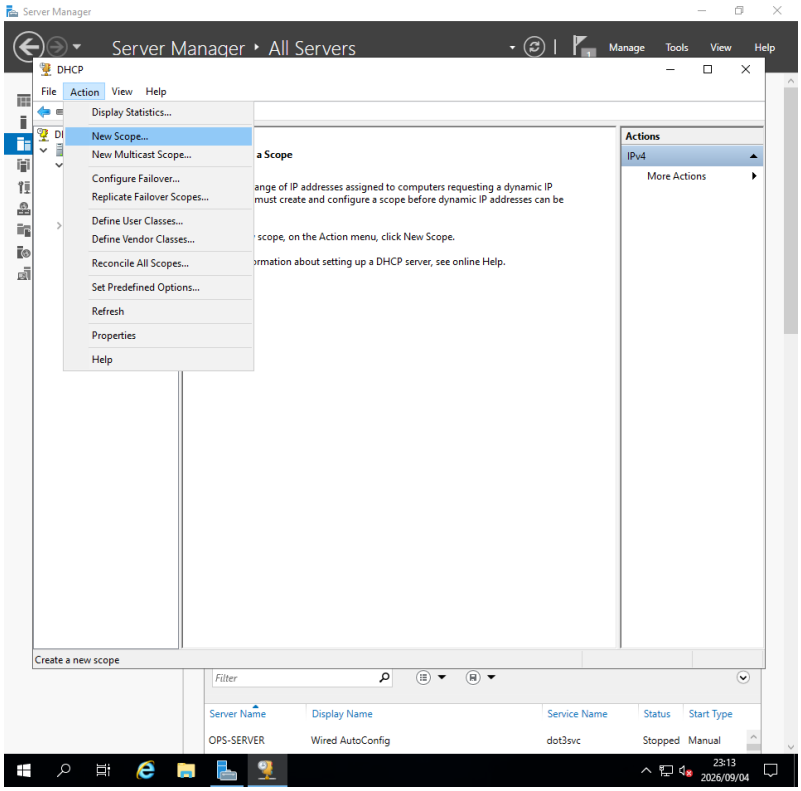


Figure 3.6: Adding a Scope

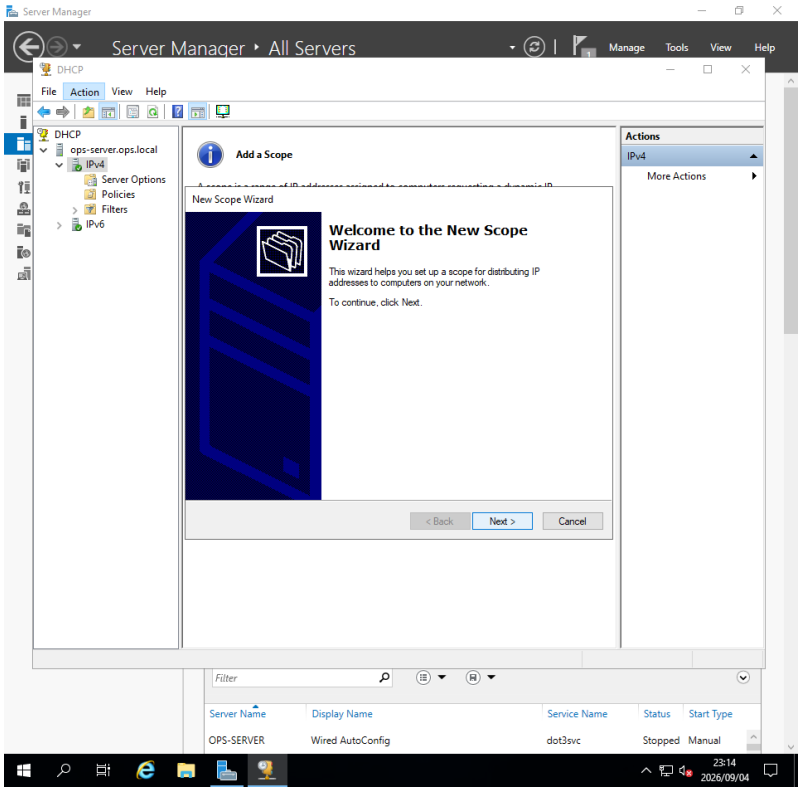


Figure 3.7: New Scope Wizard

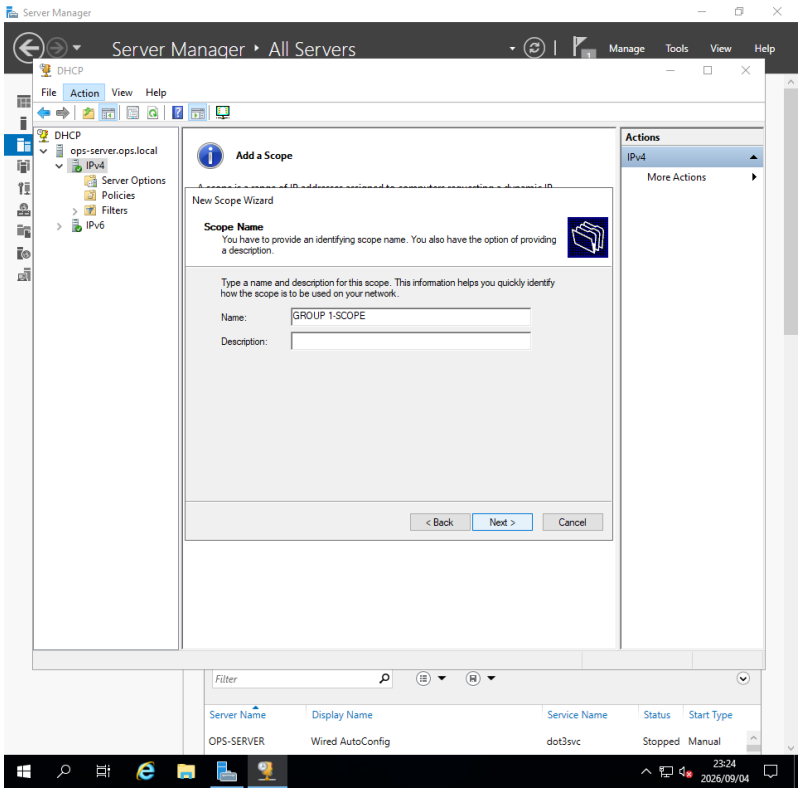


Figure 3.8: Group One Scope Made

There exists 4.3 billion posible IPv4 addresses and 3.7 billion available addresses, and therefore, there exists a countless amount of possible addresses that the scope can allow ([4]). A scope does not need to accomodate billions of addresses, it just needs a few hundred or a few dozen. When configuring it, the range of IP addresses needed to be decided. The addresses that the scope will hand out were from 192.168.0.100 to 192.168.0.254. This will be the address pool. There were no exclusions or delays made in the scope. Alongside handed out IPs, the scope also has the option to configure the amount of time that a client keeps an assigned IP before renewal, ensuring efficient management of IP addresses ([5]). This period of time is called the lease duration. It was left at the default number of 8 days.

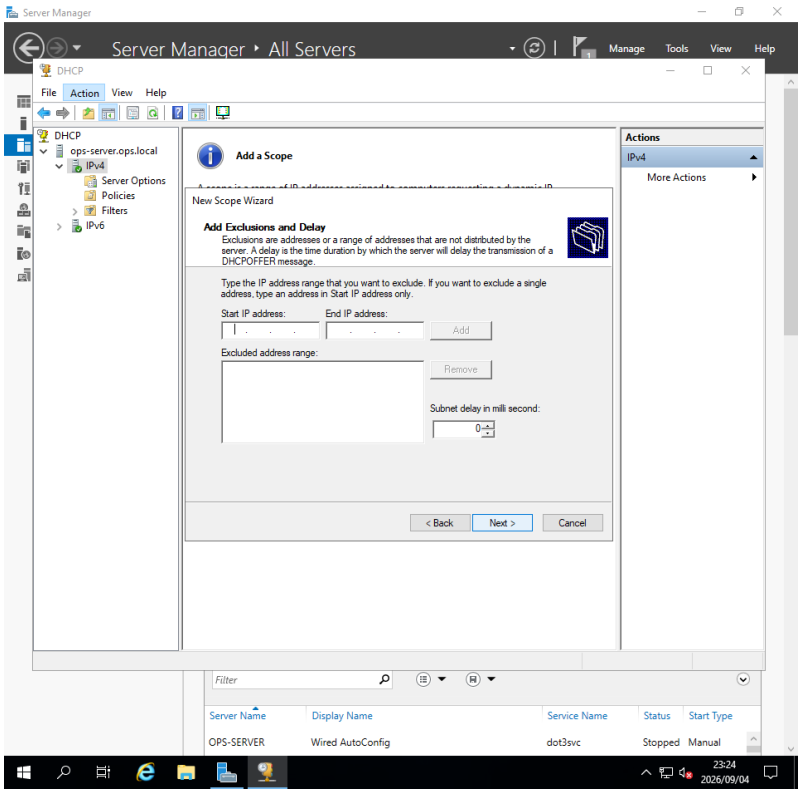


Figure 3.9: No Exclusions made

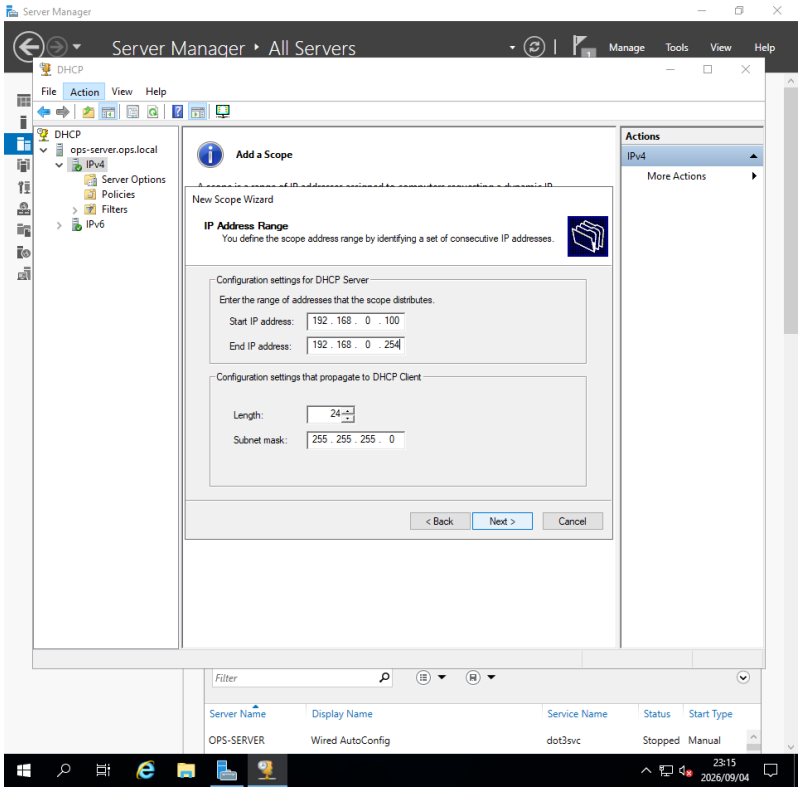


Figure 3.10: Setting IP Address Range

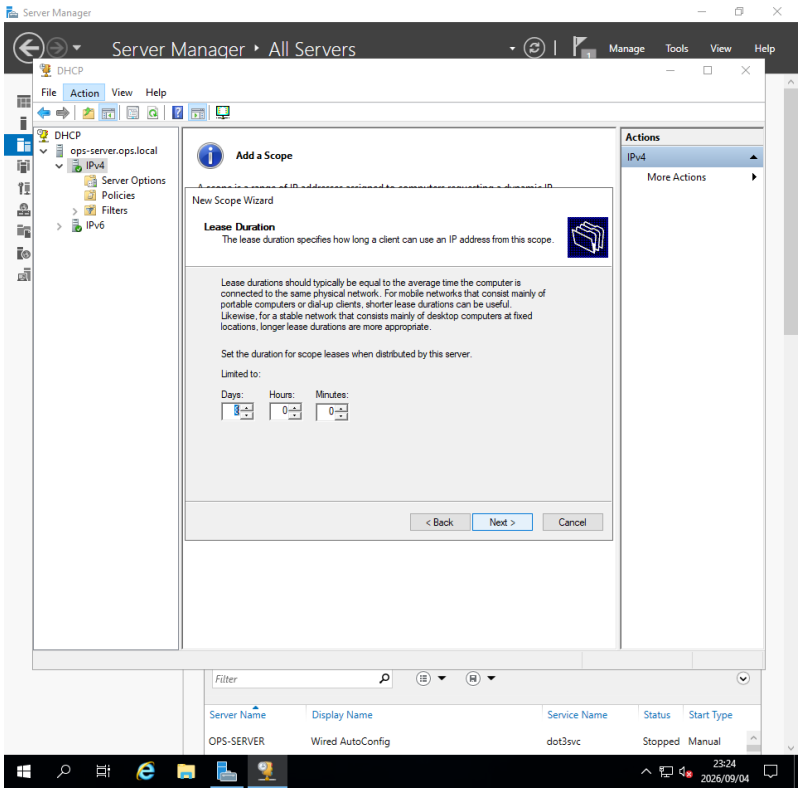


Figure 3.11: Lease duration of 8 days set

When clients use the server, access to the internet is an important feature that must be present in the server. In order for the clients to able to access the internet, they need to be able to talk to other devices and networks outside of the DHCP subnet. A default gateway allows for this. Without it, a client can only reach devices and networks inside its own subnet. The default gateway was given the IP address of 192.168.0.1.

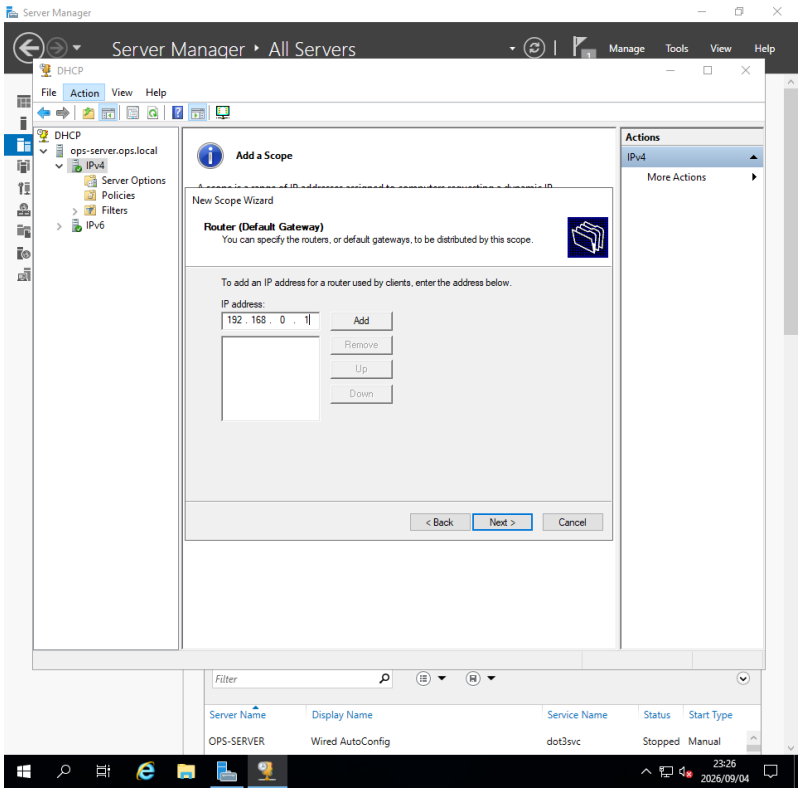


Figure 3.12: Default Gateway Configured (1)

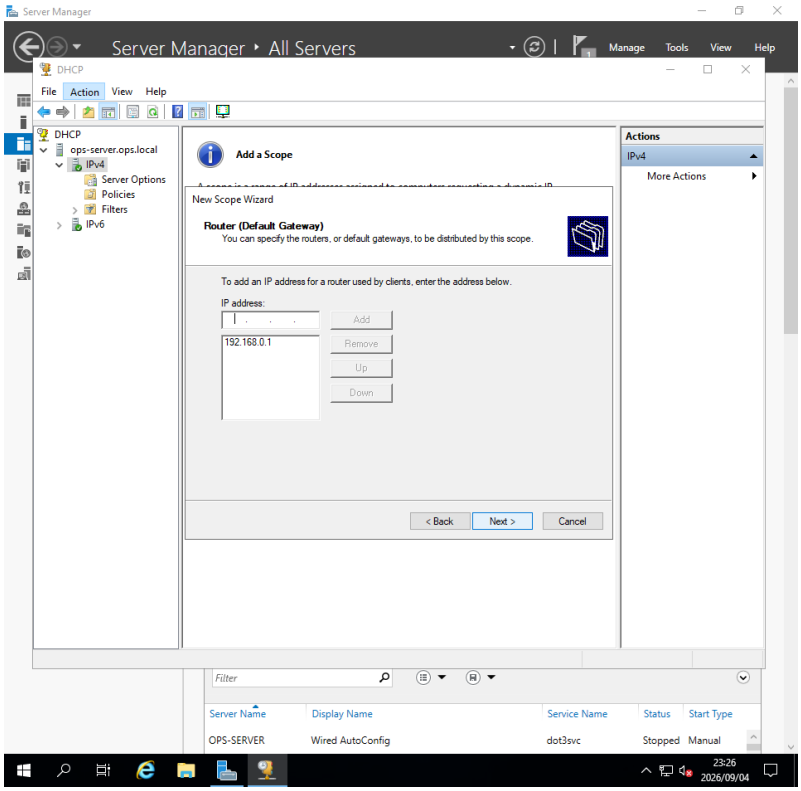


Figure 3.13: Default Gateway Configured (2)

After the setting up of the default gateway, the next step was setting up the DNS server. As mentioned earlier, the DHCP has the job of handing out IP addresses and more for a client to be able to use the network. An IP allows a device to communicate with other hosts, but IP addresses are inconvinient to memorize or type down. The DNS Server and Domain Name exist to allow for devices to access the DHCP server through use of IPs that resolve into domain names ([6]). The domain name was set to OPS.local and two IPs, 192.168.0.5 and 8.8.8.8, for two servers were set. This would mean that in order to access the server they would simply need to type in the domain name, making it more accessible for non-domain joined machines.

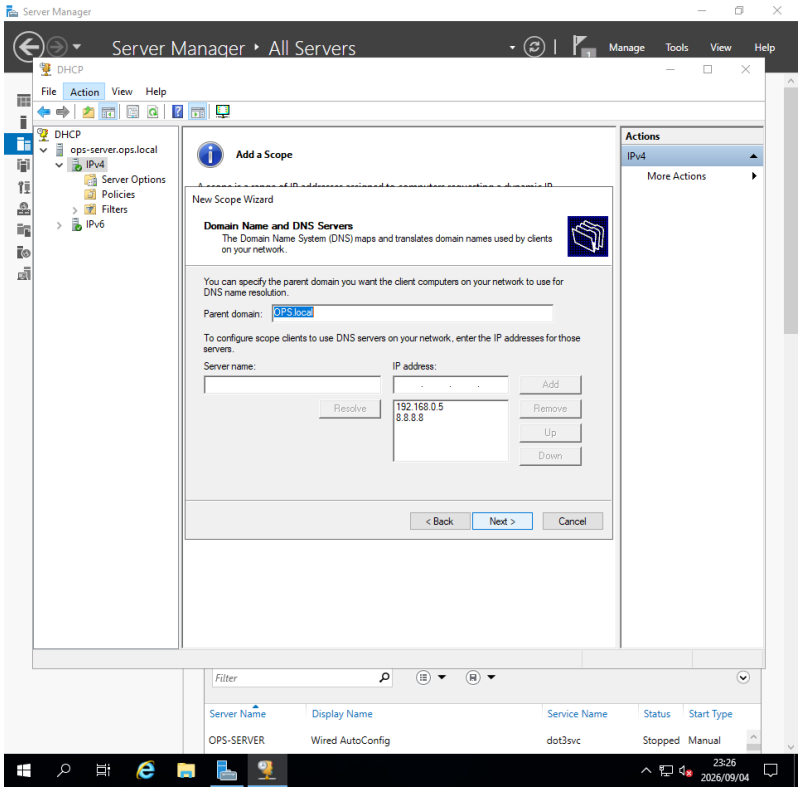


Figure 3.14: DNS Server and Domain Name

After the DNS Server and Domain Names are configured, there was an option for WINS Server configuration. WINS is a part of the scope that resolves NetBOIS computer names to IP Addresses. It is needed to support pre-Windows 2000 systems and older application using NetBOIS over TCP/IP ([7]). None of the systems planned to be connected in the network do not qualify as pre-Windows 2000 systems, there WINS is not needed.

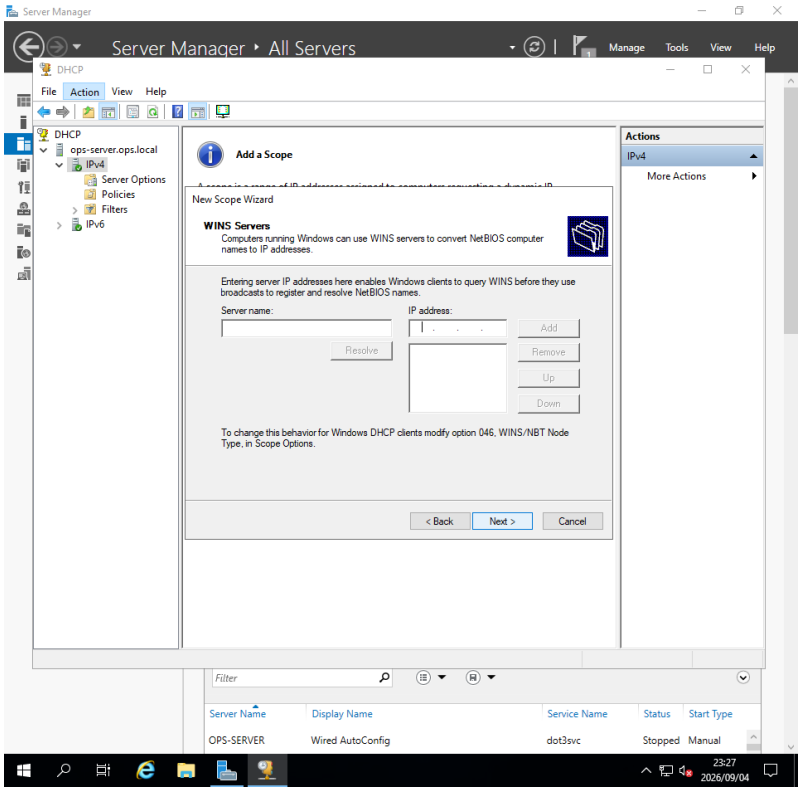


Figure 3.15: WINS Server configurinf left vacant

All necessary parts of the scope have been configured. After passing the WINS Server configuring, all that was left was activation of the scope. Upon activating the scope, the wizard will present completion display. To close the wizard, the finish button was clicked. The scope is now activated and ready for testing. The scope will now had out IP addresses from 192.168.0.100 to 192.168.0.254 and this will be its address pool. At the time of the creation of the scope, there were no address leases yet.

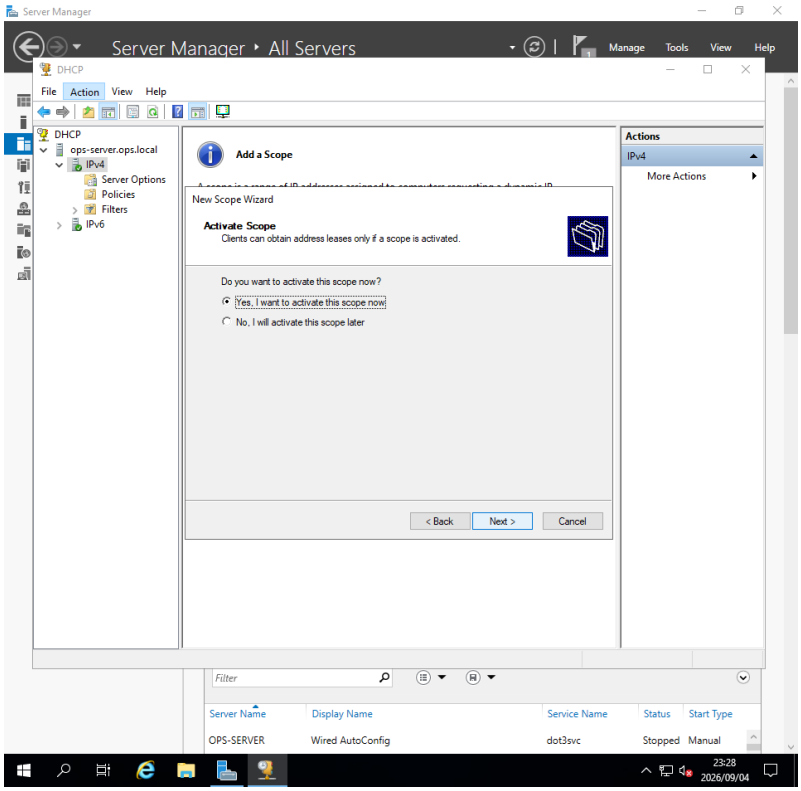


Figure 3.16: Activation of Scope

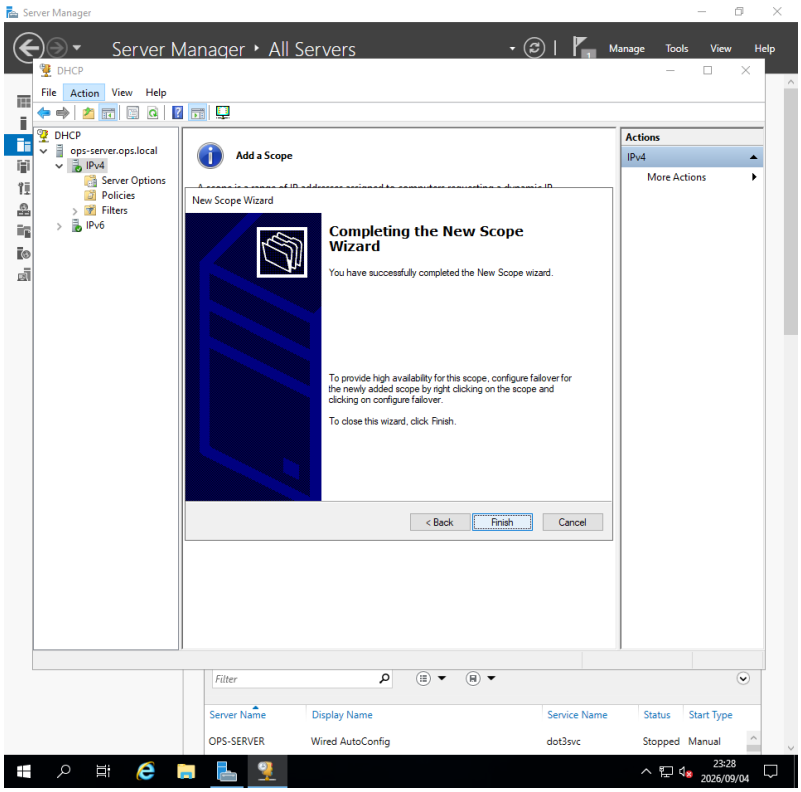


Figure 3.17: Finishing New Scope Wizard

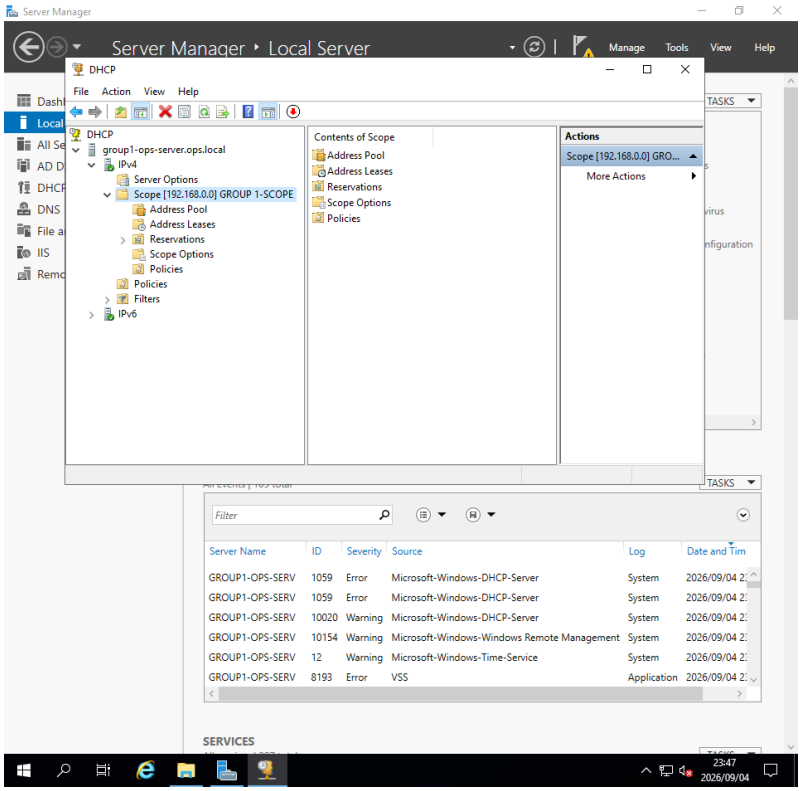


Figure 3.18: New DHCP Scope created

3.3 Issues encountered for DHCP Server Setup and Alternative Paths

3.3.1 Use of Powershell in event of DHCP Window Failure

On some occassions, the DHCP console can stop functioning due to either a domain name change or a new ethernet being added to it for internet access. This means that configuring a new scope using the window is not possible, however, this does not mean that new DHCP scopes cannot be configured or removed. Windows Powershell is still available on Windows Server 2019 and it serves as a way to bypass any surface level server issues. Using powershell commands allow you to still configure and modify DHCP scopes with a similar degree as if you were using the console. They can also be used when you simply prefer Powershell.

Some of the important commands you would need to configure DHCP IPv4 scopes with Powershell are as follows ([8]):

```
Add-DhcpServerv4Scope - adding scopes
Get-DhcpServerv4Scope - viewing scopes
Set-DhcpServerv4Scope - modify scopes
```

```
Remove-DhcpServerv4Scope - delete scopes
Get-DhcpServerv4ScopeStatistics - show usage statistics
Add-DhcpServerv4Lease - add lease
Get-DhcpServerv4Lease - view leases
Remove-DhcpServerv4Lease - remove Lease
Add-DhcpServerv4Reservation - add reservations
Get-DhcpServerv4Reservation - view reservations
Set-DhcpServerv4Reservation - modify reservations
Remove-DhcpServerv4Reservation - delete reservation
```

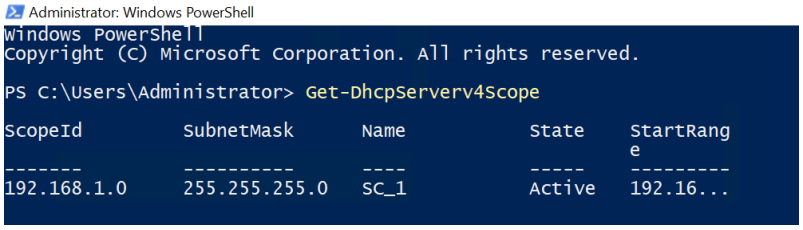


Figure 3.19: Example of accessing DHCP through Powershell

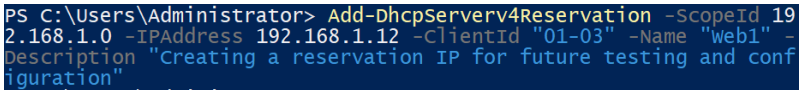


Figure 3.20: Example of adding a DHCP reservation



Figure 3.21: Example of verification of DHCP reservation

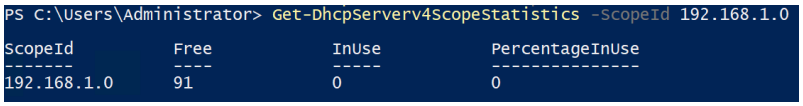


Figure 3.22: Example of IP address tracking

3.3.2 Use Windows Powershell to create DHCP scope

To create a DHCP scope using Windows Powershell, it must be opened as administrator([9]). Considering that scopes have most likely been made already, a command must be written to check for existing scopes.

The command Get-DhcpServerv4Scope will list every scope in the DHCP server that was created using the DHCP console ([9]). If there wasn't any preexisting scopes, nothing will show.

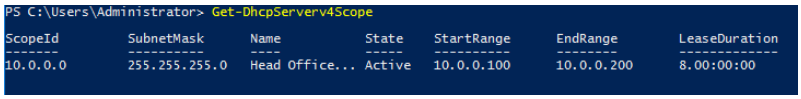


Figure 3.23: Example of Get-DhcpServerv4Scope command in use

To create scope in Powershell normally would involve a lot of typing, as the Add-DhcpServerv4Scope command has a lot of parameters. A conventional use of the full command would look like the following:

```
Add-DhcpServerv4Scope `
    -Name "Test Scope" `
    -Description "Scope for DHCP testing" `
    -StartRange 192.168.10.100 `
    -EndRange 192.168.10.200 `
    -SubnetMask 255.255.255.0 `
    -LeaseDuration 0.08:00:00 `
    -State Active `
    -PassThru
```

This can be tedious, so splatting can be used to make passing parameters easier([10]). Powershell Splatting is when you use a hashtable to create a parameter set that is passed to the cmdlet. The cmdlet will create DHCP scope according to the given parameters. For example, if you entered the following in the powershell console:

```
$HashArgs = @{
     'Name' = 'Head Office Wireless Scope';
     # This sets the name of the wireless scope
     'Description' = 'Wireless Clients';
     # This is the human-readable description of the scope
     'StartRange' = '10.1.0.100';
     # Specifies the starting IP address in the scope
     'EndRange' = '10.1.0.200';
     # Specifies the end IP address in the scope
     'SubnetMask' = '255.255.255.0';
     # Specifies the subnet mask of the scope
     'State' = 'Active';
     # Activates the scope
     'LeaseDuration' = '1.00:00:00';
     # Specifies the length of the lease duration
 }
 Add-DhcpServerv4Scope @HashArgs
```

The hashtable @HashArgs is given DHCP server parameters When the Add-DhcpServerv4Scope command is entered, a new scope named Head Office Wireless Scope is made, with an address pool between 10.1.0.100 and 10.1.0.200, with a subnet of 255.255.255.0([9]). The lease duration is a day and the scope is made active. To check if the scope was made, run Get-DhcpServerv4Scope again. A way to make looking at the scopes easier is to add " | Format-List" at the end of the command([9]).

3.4 Testing

3.4.1 Checking Server

To test the operation of a DHCP server, there must be confirmation that data packets are travelling between the server and client devices. To start, we must check the server on a Powershell window that is run as administrator. We must run the command:

```
Get-Service -Name (Name of DHCP server)
```

The status collumn must display running. If the command shows the service as Stopped, that means that the server can't answer any client requests. The reason behind that issue, might be that the server somehow hasn't started or the server hasn't been authorized. If it needs to be started, run the command

```
Start-Service (Name of DHCP server)
```

after that running the get service command will help confirm if it running. Another probable reason is that the server needs authorization. The command

```
Get-DhcpServerInDC
```

will verify if the DHCP server is authorized in Active Directory([11]). However for this to work, you need to first add the server into the Directory with the command

```
Add-DhcpServerInDC -DnsName (name of server)
-IPAddress (address of server)
```

A few other checks to do are to check the configuration and the scope usage, seeing howing addresses are in use and how many are free, with the following commands([11]).

```
Get-DhcpServerv4ScopeStatistics
Get-DhcpServerv4Scope
```

3.4.2 Test from a client and Verifying on server

To test a server from a client, it will have to request a lease. To request a lease on a Windows client, you will have use Command Prompt and run ipconfig /release, then ipconfig /renew, then ipconfig /all. ipconfig /release releases a DHCP client lease, ipconfig /renew renews the client lease, ipconfig /all or just ipconfig verifies a DHCP client lease([12]).

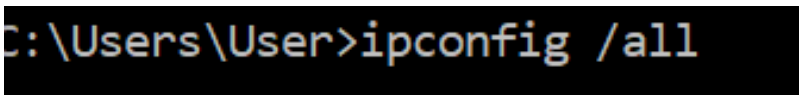


Figure 3.24: ipconfig /all command in Command prompt

The address, subnet mask, gateway, DNS servers and DHCP server line must all match your scope.

There is a chance that your client will have incorrect or missing options present. The cause of this could be that the network configuration isn't working correctly. This is due to the server clients not being able to support the missing options, but it also could be possible that the server isn't configured to hand them out ([13]).

The solution to this is to change the IP address list for the missing or incorrect options at the DHCP scope in use. You might have to configure the options as a Server Option at the DHCP server, remove it and set the correct value in the Scope Options for the DHCP scope that is used by the cleint([13]).

After the lease is recorded, run the command Get-DhcpServerv4Lease -ScopeId <scope> to confirm clients are receiving addresses. The cmdlet also has a means of identifying bad leases with -BadLeases and -AllLeases to specify all IPv4 address leases([14]). The logs of the server and leases are stored in an Event Viewer and it tracks major DHCP service events, but individual lease events go to seperate DHCP log files. By default, they are in the directory `C:\Windows\system32\dhcp` with one file per weekday([15]).

4. Group Policy setup and configuration for Windows Server 2019

4.1 Introduction

Group Policy is a feature of Microsoft Windows that provides centralized management and configuration of operating systems, applications, and users' settings in an Active Directory environment. It allows administrators to define security settings, software installation policies, and user interface configurations across multiple computers within a domain.

Well implemented Group Policy can reduce risk, improve security, and ensure consistency across the network. Group Policy Objects (GPOs) can be define for specific users, groups, or computers, and can be linked to organizational units (OUs) to apply settings based on the structure of the Active Directory.

This chapter will cover the setup and configuration of Group Policy in a Windows Server 2019 environment, including the creation of GPOs, linking them to organizational units (OUs), and testing their application.

4.2 How does Group Policy work?

Group Policy relies on the Active Directory infrastructure to apply settings to users and computers. When a user logs in or a computer starts up, the Group Policy engine retrieves the relevant GPOs from the domain controller and applies the settings defined within those GPOs. The order of precedence for applying GPOs is as follows:

1. Local Group Policy: Settings defined on the local computer.
2. Site-level GPOs: Settings defined for the site in which the computer resides.
3. Domain-level GPOs: Settings defined for the entire domain.
4. Organizational Unit (OU)-level GPOs: Settings defined for the specific OU in which the user or computer resides.

When multiple GPOs are applied, the settings from the highest precedence GPO will override those from lower precedence GPOs. This allows administrators to create a hierarchy of policies that can be tailored to specific users, groups, or computers.

Policies are refreshed periodically (every 90 minutes by default), and administrators can also force a Group Policy update using the ‘gpupdate‘ command. This ensures that any changes made to GPOs are applied immediately.

4.3 Assignment specified Group Policy configuration

In this scetion we will cover the specific Group Policy configuration required for the assignment. The policies required include disabling command prompt, restricting removable storage access, preventing software installation, and enforcing password complexity requirements. Each of these policies will be configured through the Group Policy Management Console (GPMC) and linked to the appropriate organizational units (OUs) within the Active Directory.

The sections that follows will provide step-by-step instructions for creating and configuring the necessary GPOs, as well as testing their application to ensure they are functioning as intended.

4.4 Organizational Unit creation and configuration

The workflow for creating and configuring Group Policy Objects (GPOs) involves several steps, including accessing the Group Policy Management Console (GPMC), creating a new GPO, configuring the desired settings, and linking the GPO to the appropriate organizational unit (OU). Therefore the OU structure must be defined in Active Directory before creating and linking GPOs.

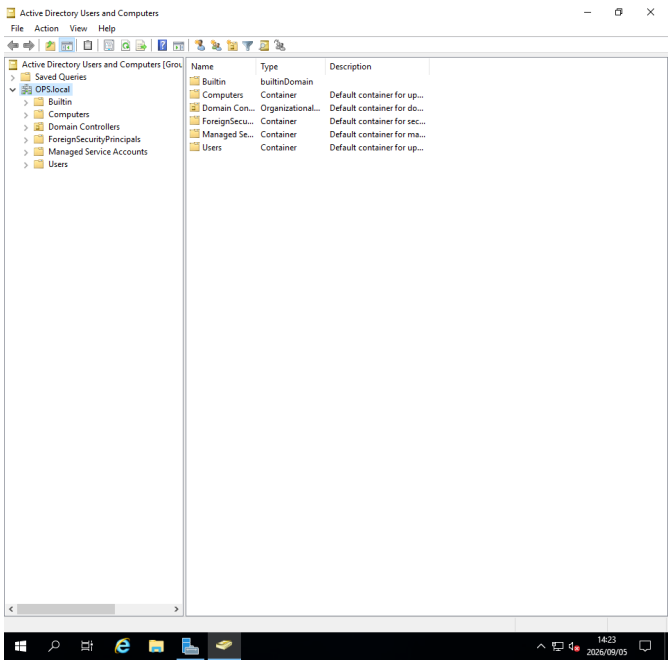


Figure 4.1: Active Directory Users and Computers.

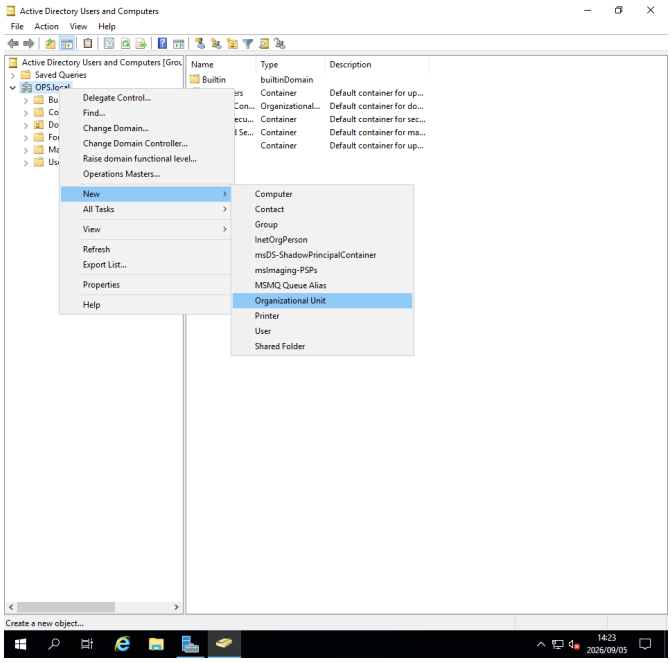


Figure 4.2: New OU object selected under the domain OPS.local.

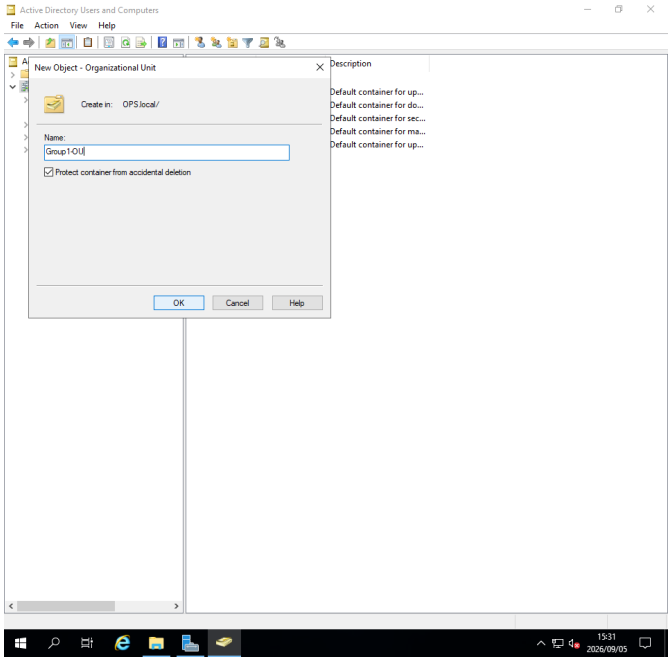


Figure 4.3: New OU object created and named Group1-OU.

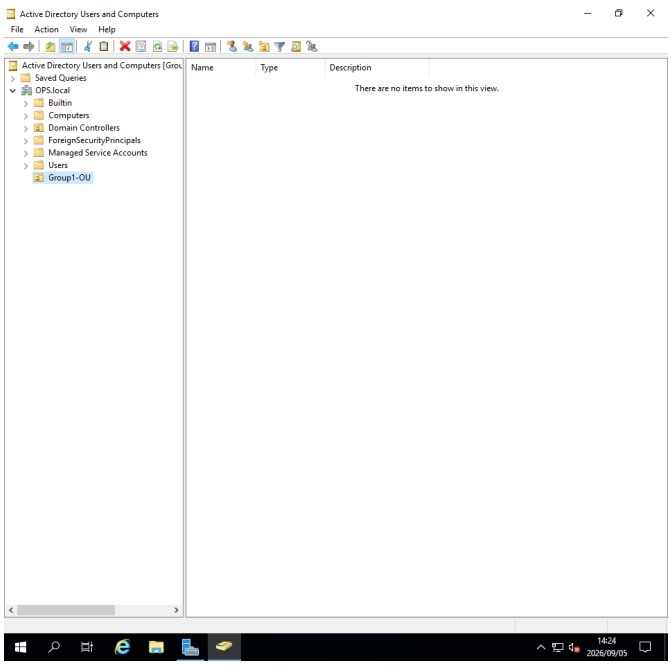


Figure 4.4: New OU object listed under the selected domain.

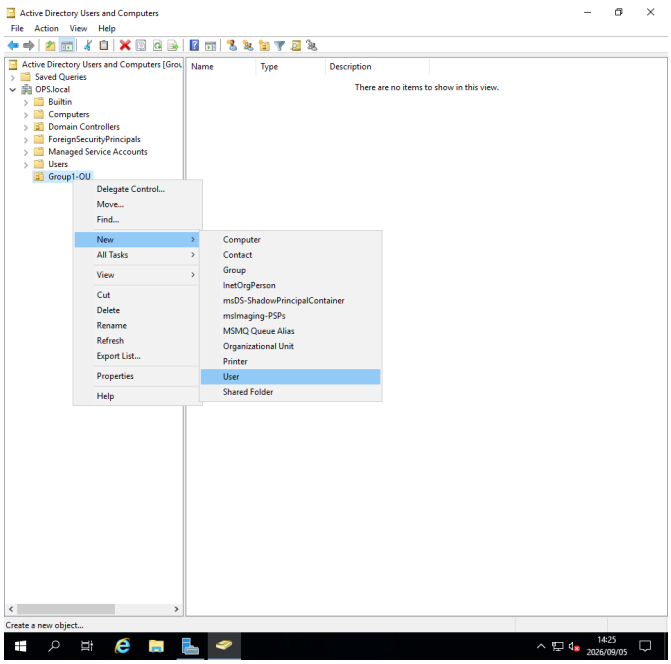


Figure 4.5: New User object selected under the Group1-OU.

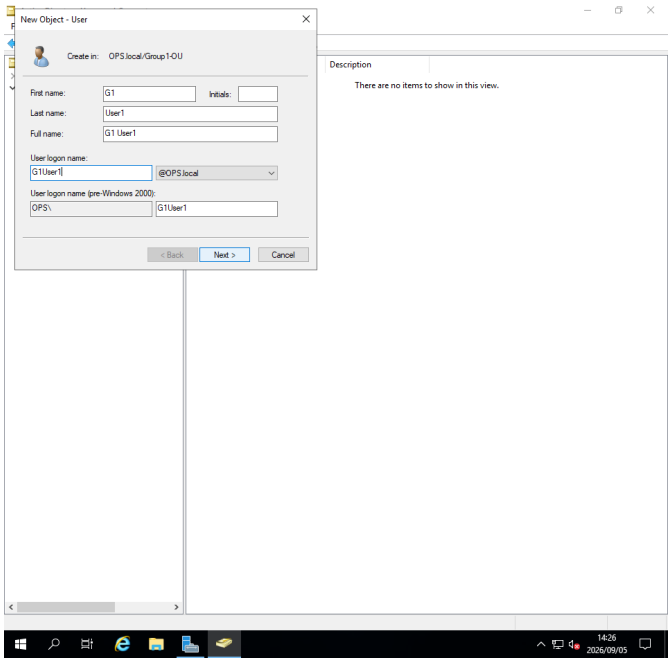


Figure 4.6: New User object configuration.

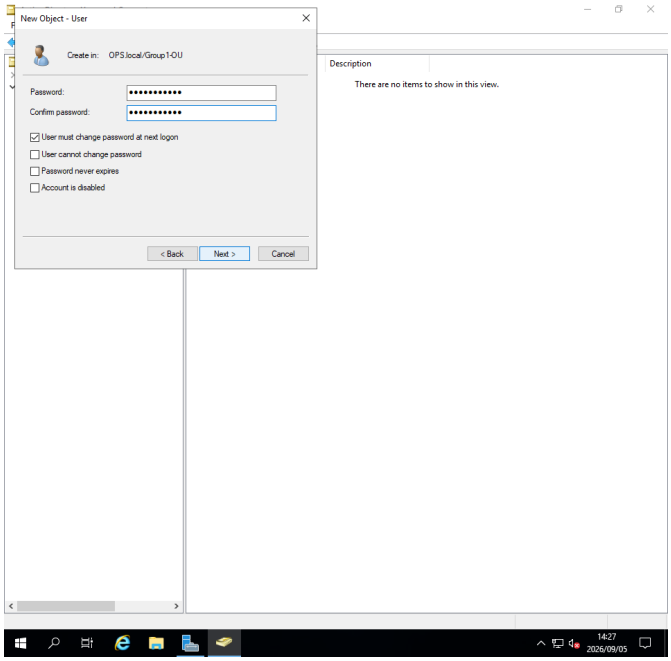


Figure 4.7: New User object password configuration.

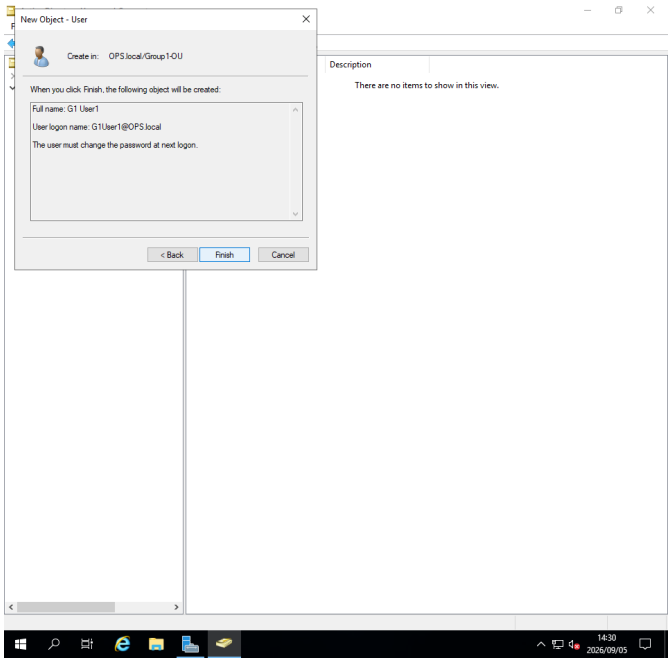


Figure 4.8: New User object summary.

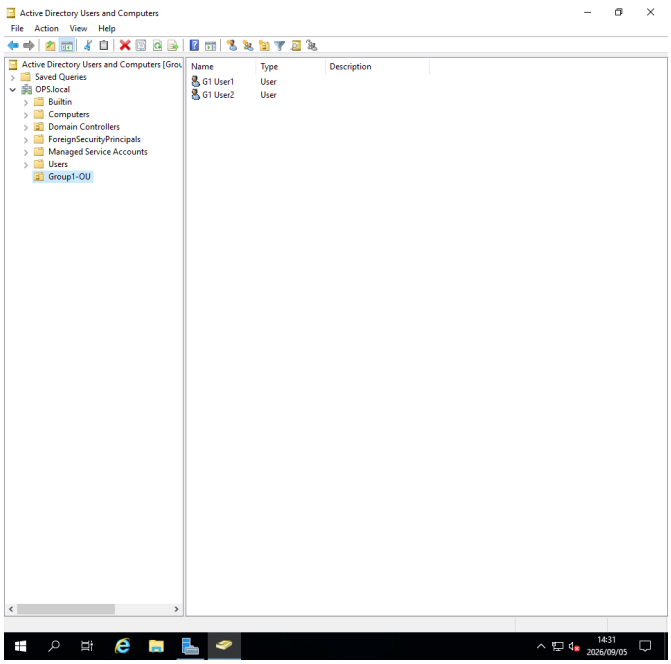


Figure 4.9: User object created and named. Here are two users listed.

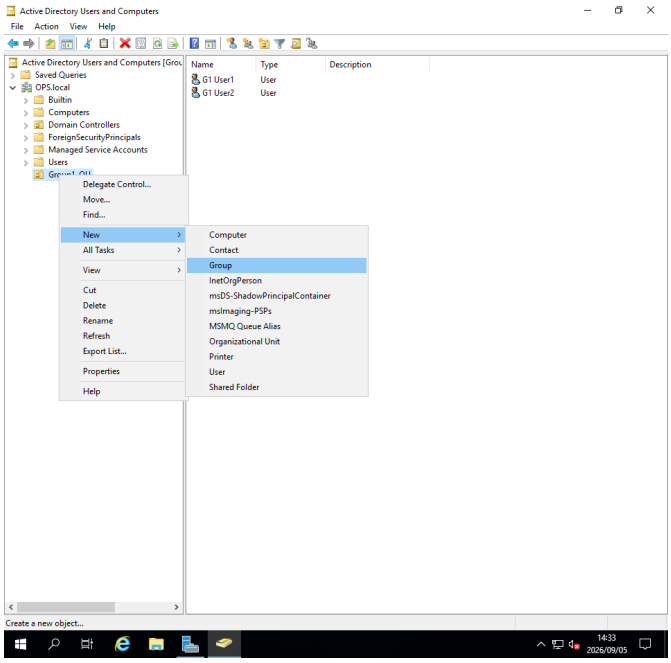


Figure 4.10: New Group Object selected under the Group1-OU. This allows the grouping of users.

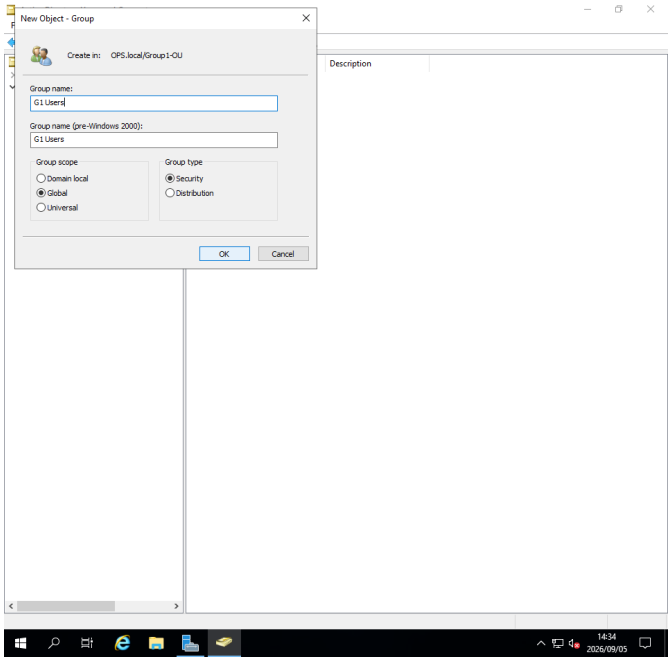


Figure 4.11: New Group object configuration. Group name, scope and type configured.

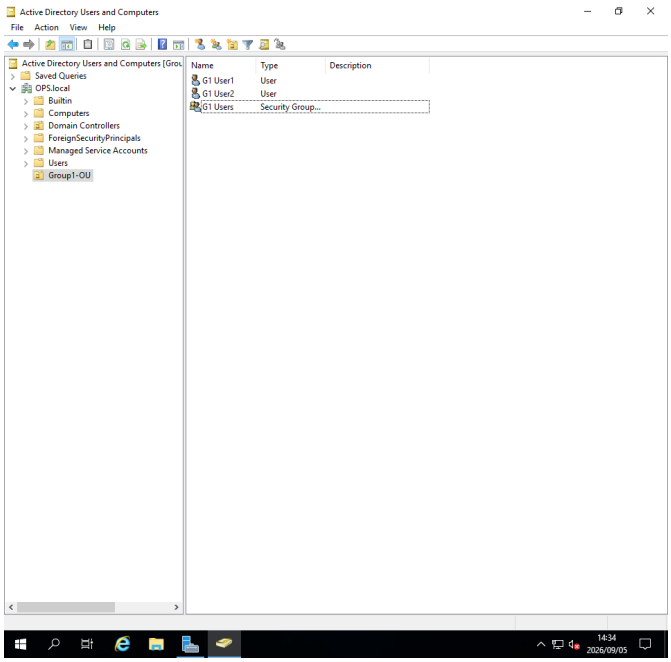


Figure 4.12: New Group object listed under the Group1-OU.

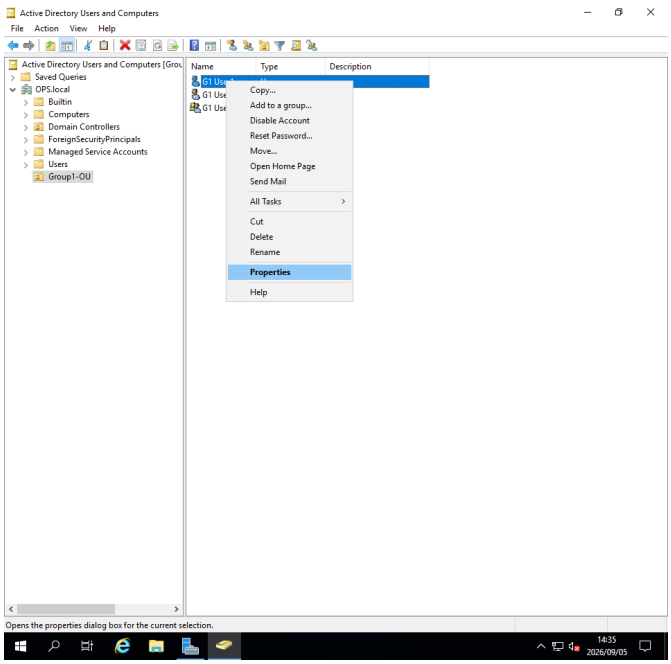


Figure 4.13: User1 showing membership configuration method 1.

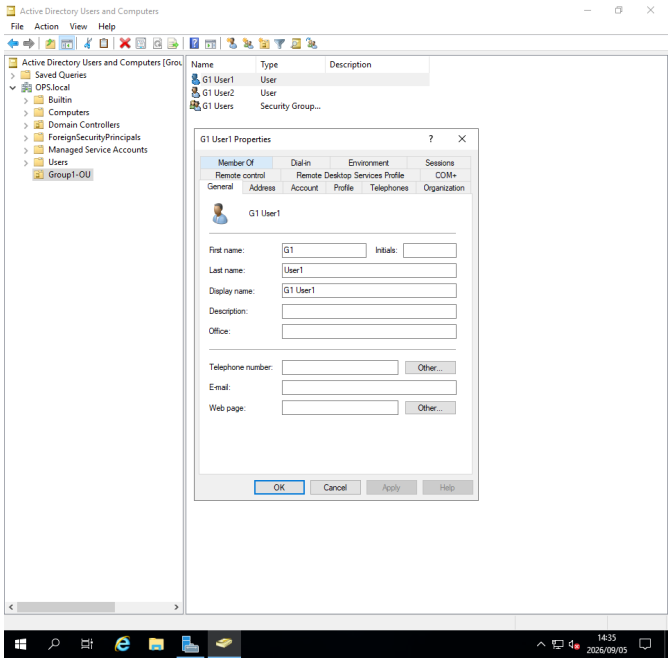


Figure 4.14: User1 membership configuration step 1.

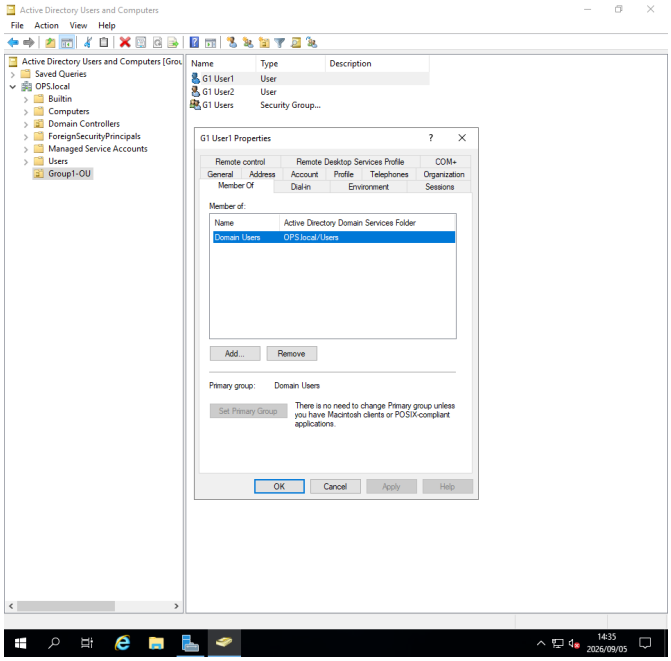


Figure 4.15: User1 membership configuration step 2.

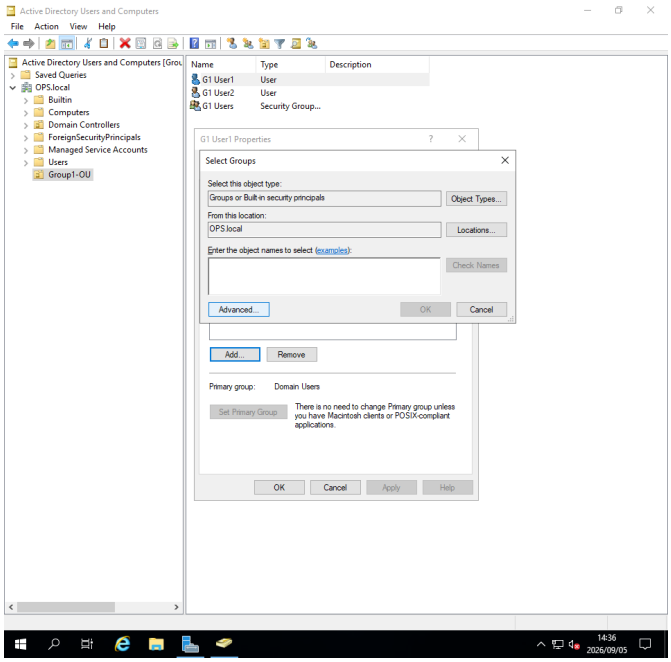


Figure 4.16: User1 membership configuration step 3.

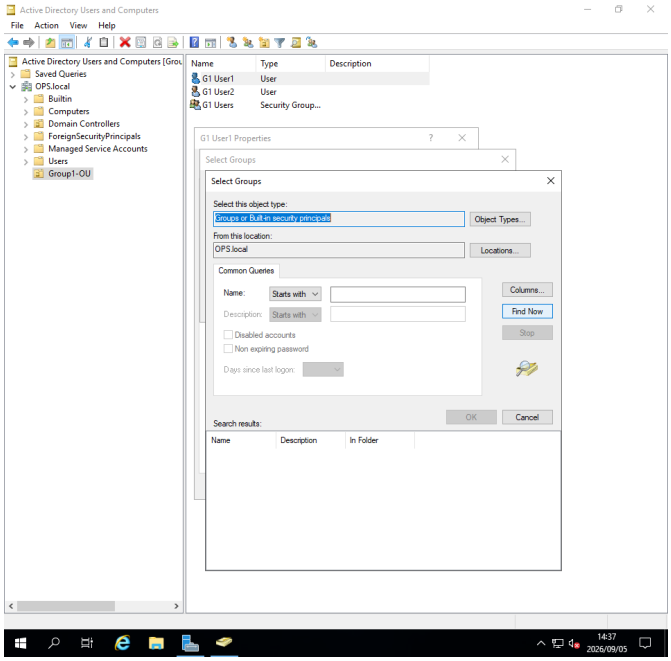


Figure 4.17: Group search window

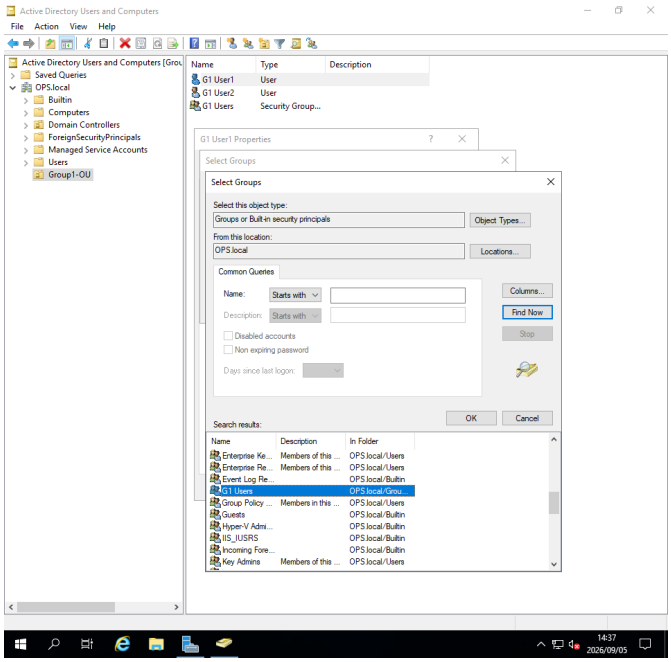


Figure 4.18: G1 Users Group find and selection.

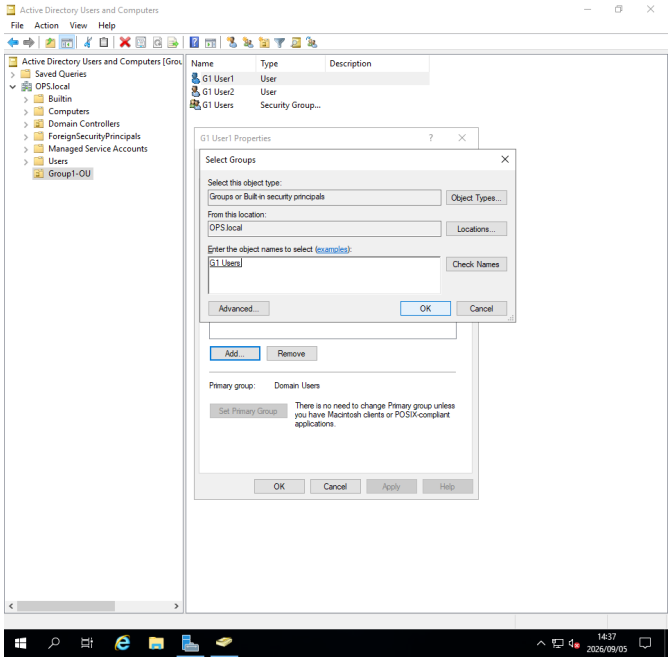


Figure 4.19: Group selection and addition window.

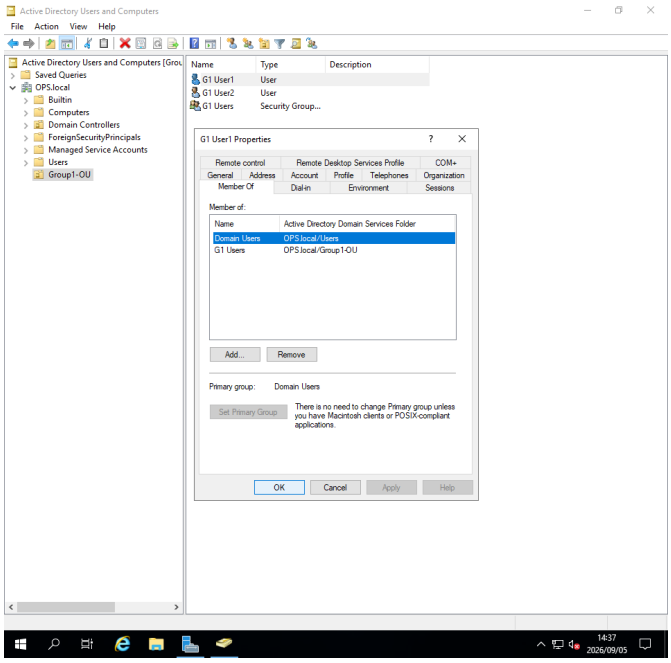


Figure 4.20: User1 membership tab.

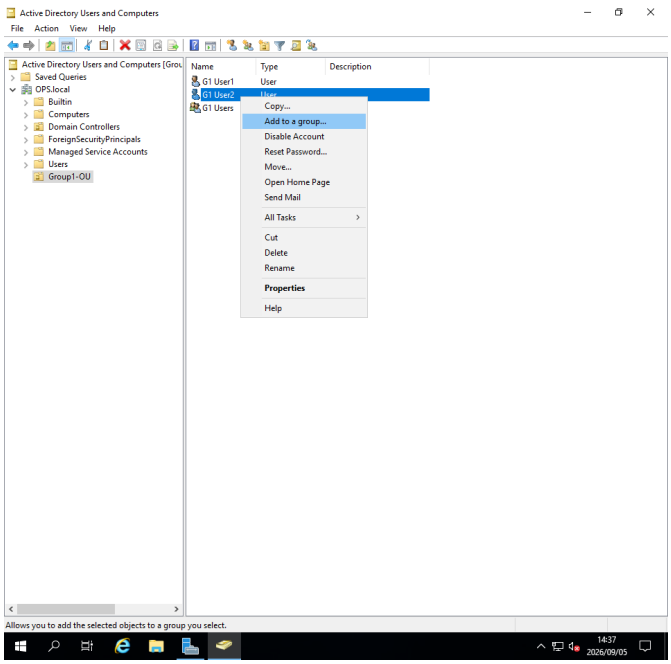


Figure 4.21: User2 showing membership configuration method 2. Add to a group selected.

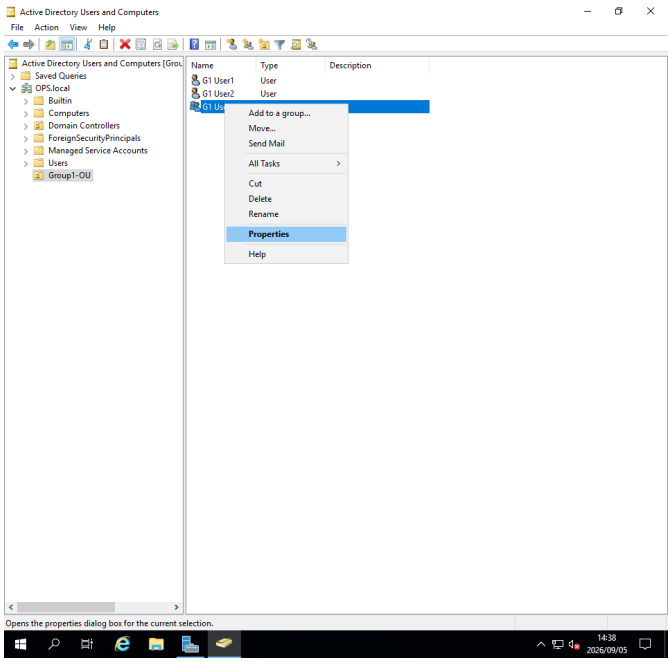


Figure 4.22: G1 Users Group properties selected to add Users to itself.

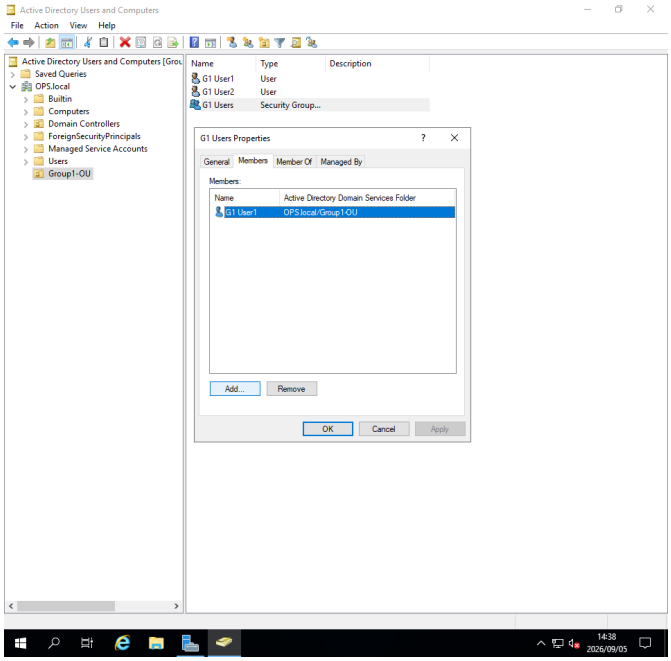


Figure 4.23: Properties tab with list of members for G1 Users Group.

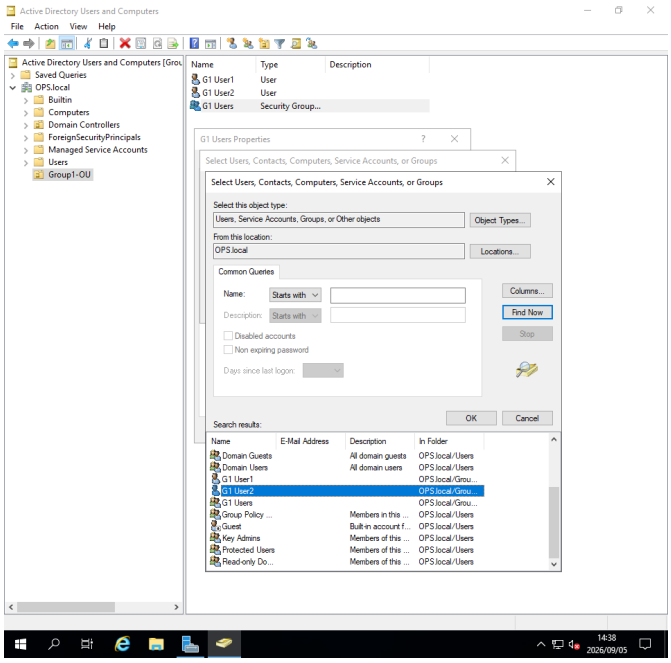


Figure 4.24: Users find and selection.

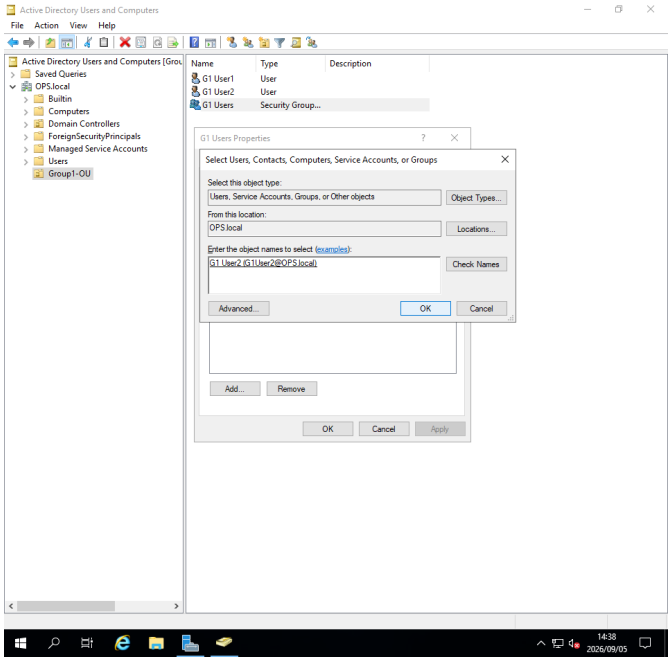


Figure 4.25: User2 selected and will be added to the group.

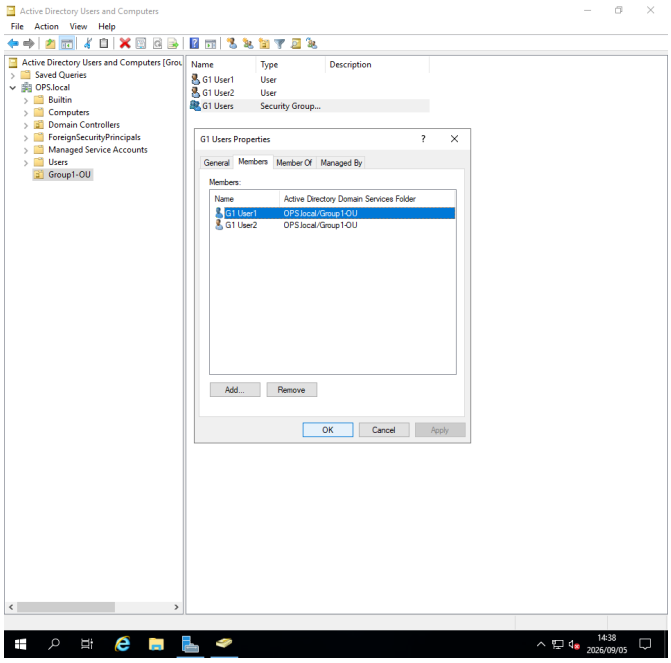


Figure 4.26: Properties tab with list of members for G1 Users Group. The list now includes User2.

4.5 Group Policy Object creation and configuration

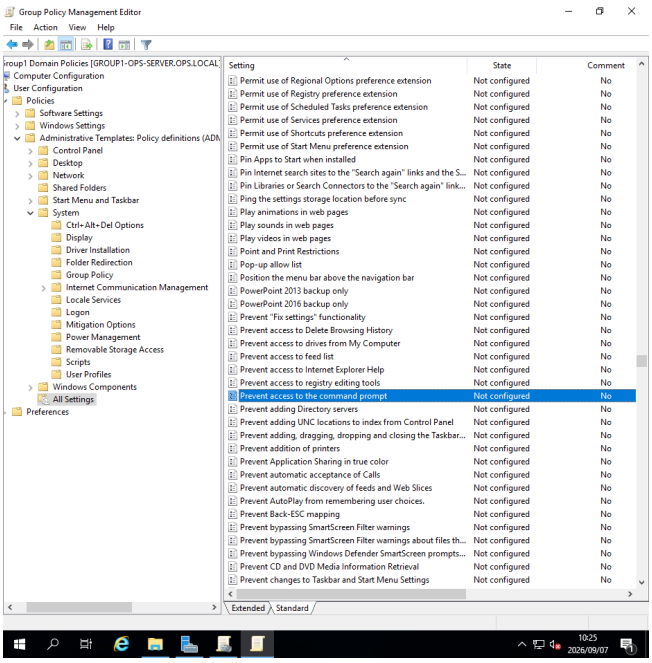


Figure 4.27: Group Policy Management Console. Prevent access to the Command Prompt selected.

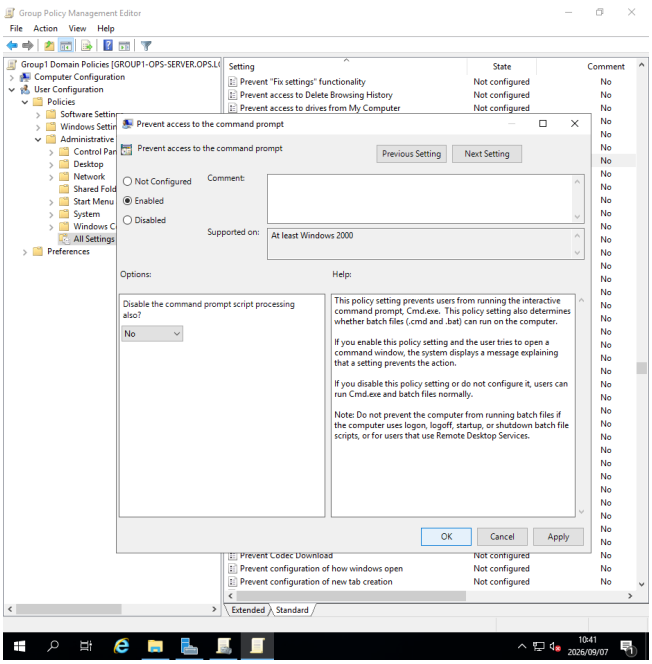


Figure 4.28: Policy settings for Prevent access to the Command Prompt. Enabled is selected.

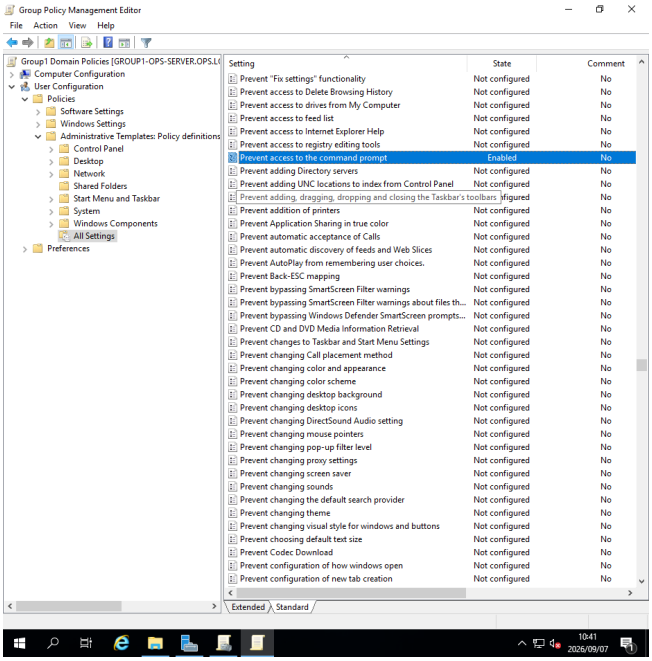


Figure 4.29: Policy settings for Prevent access to the Command Prompt shown as enabled.

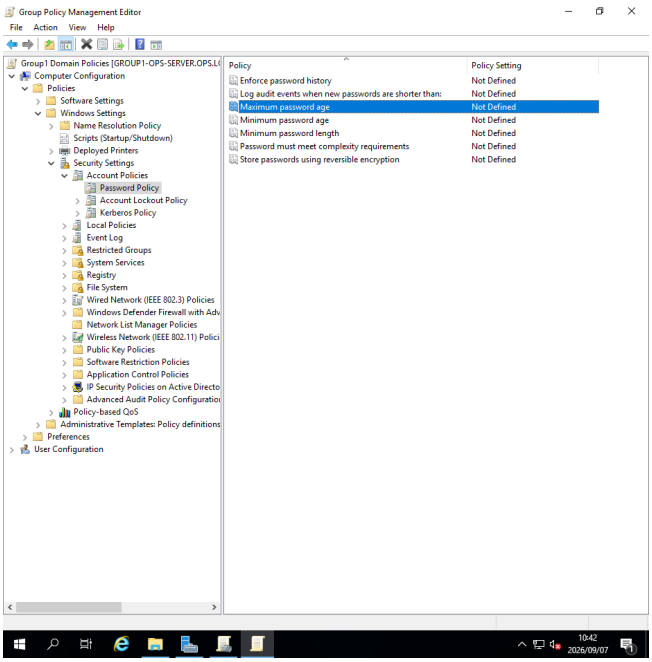


Figure 4.30: Password Policy settings. Maximum password age selected.

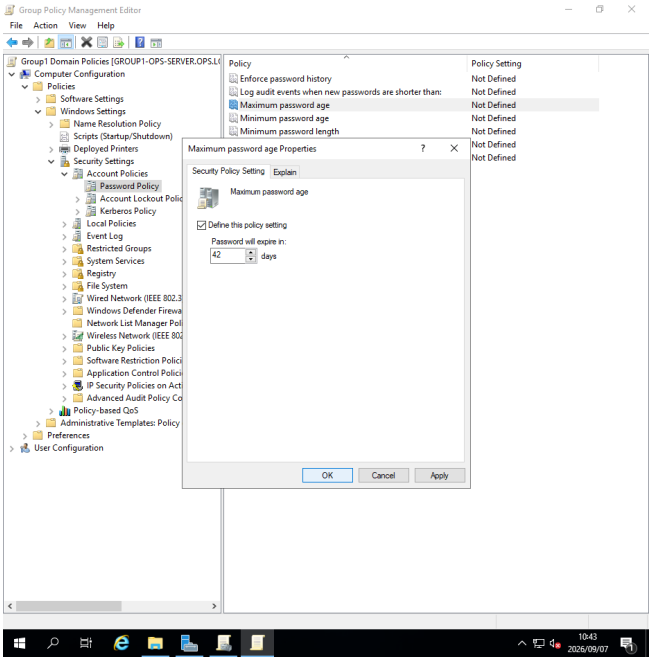


Figure 4.31: Maximum password age set to 42 days.

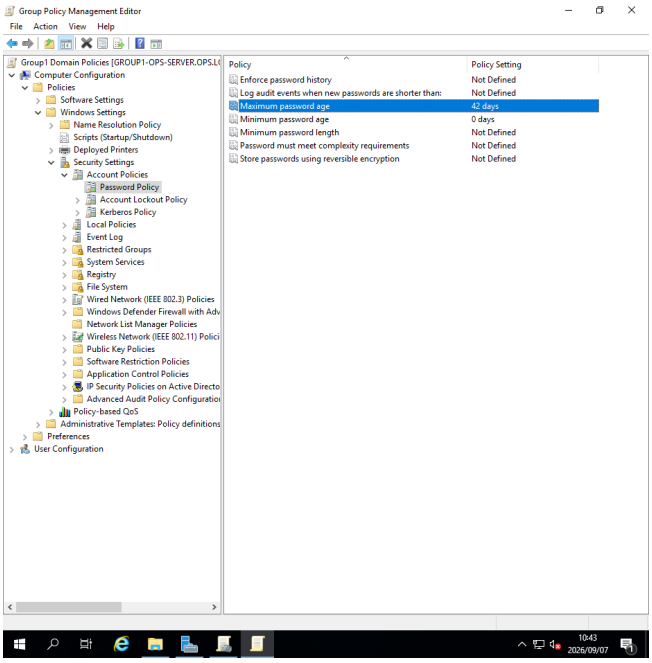


Figure 4.32: Maximum password age shown as 42 days.

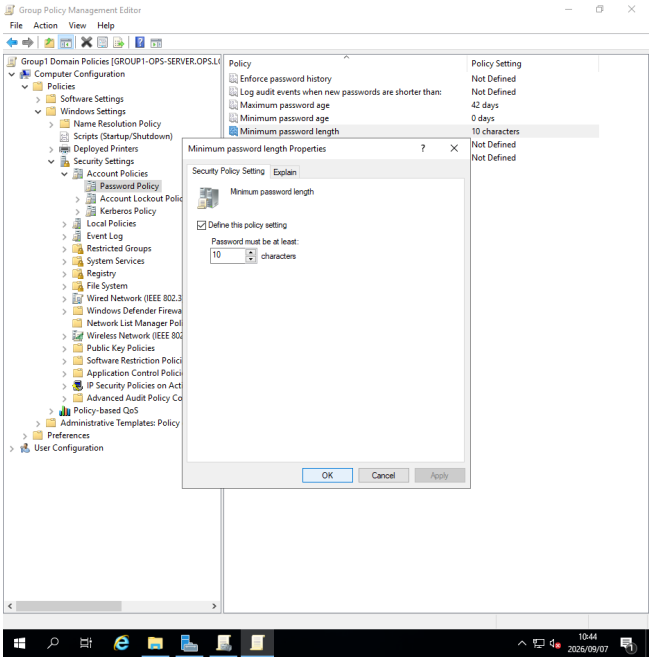


Figure 4.33: Minimum password length set to 10 characters.

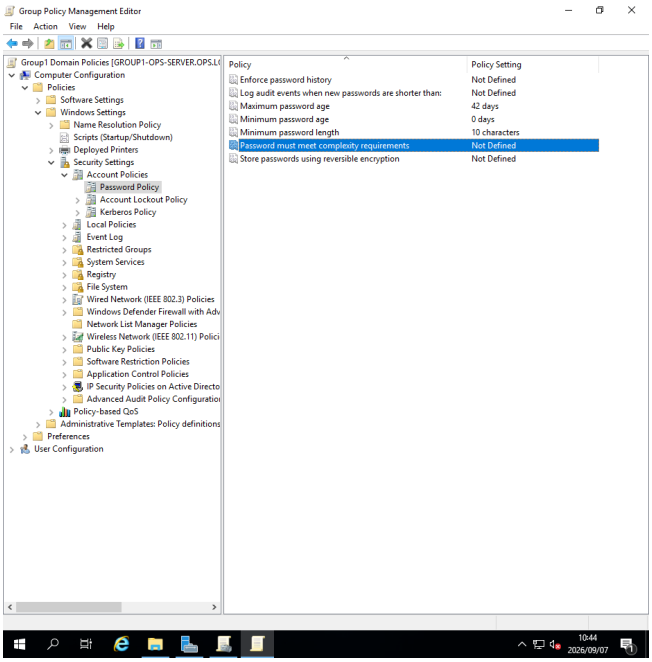


Figure 4.34: Minimum password length shown as 10 characters.

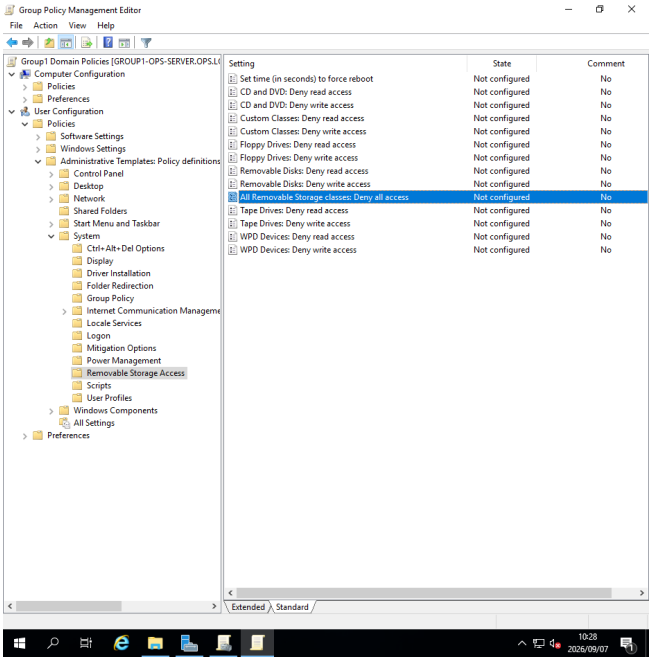


Figure 4.35: All removable storage classes: Deny all access selected.

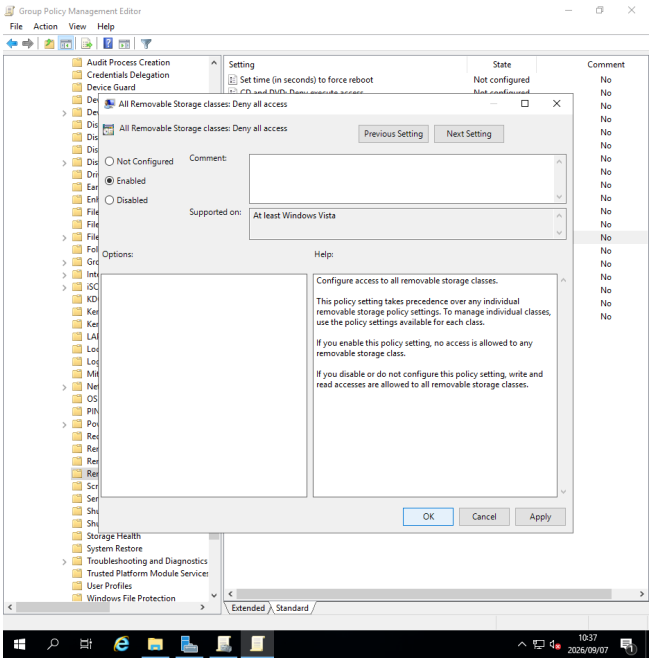


Figure 4.36: All removable storage classes: Deny all access Enabled.

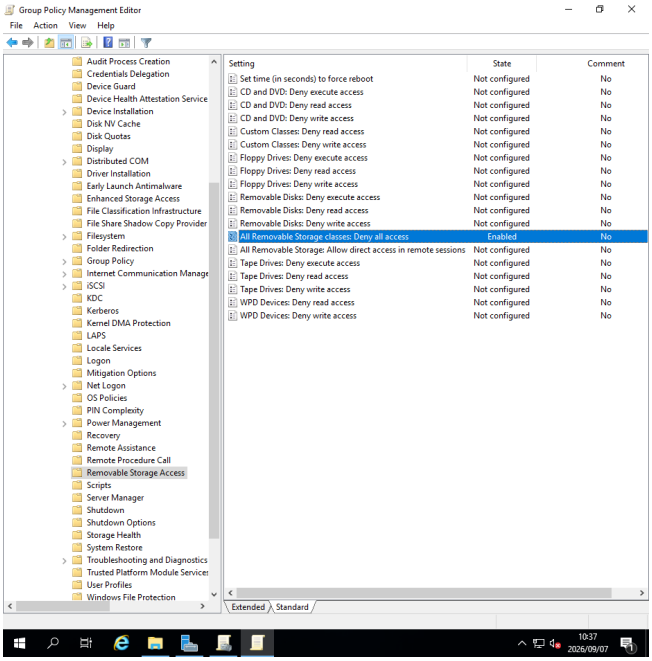


Figure 4.37: All removable storage classes: Deny all access shown as enabled.

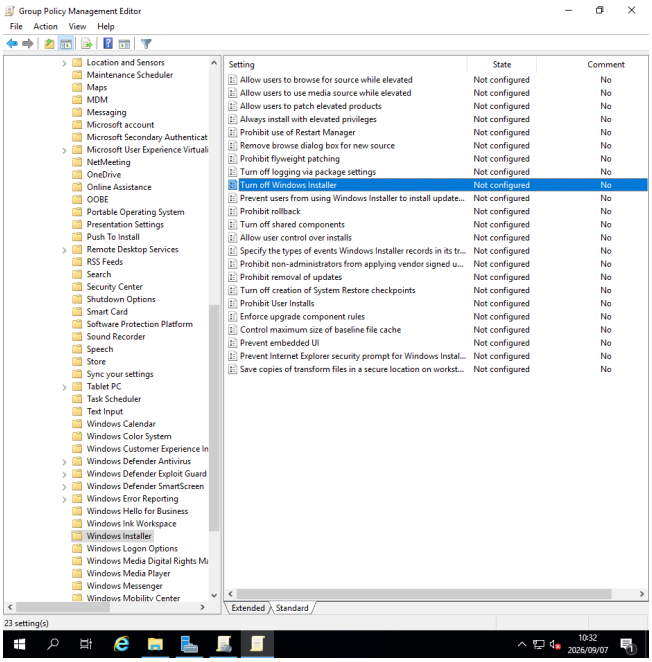


Figure 4.38: Turn off Windows Installer selected.

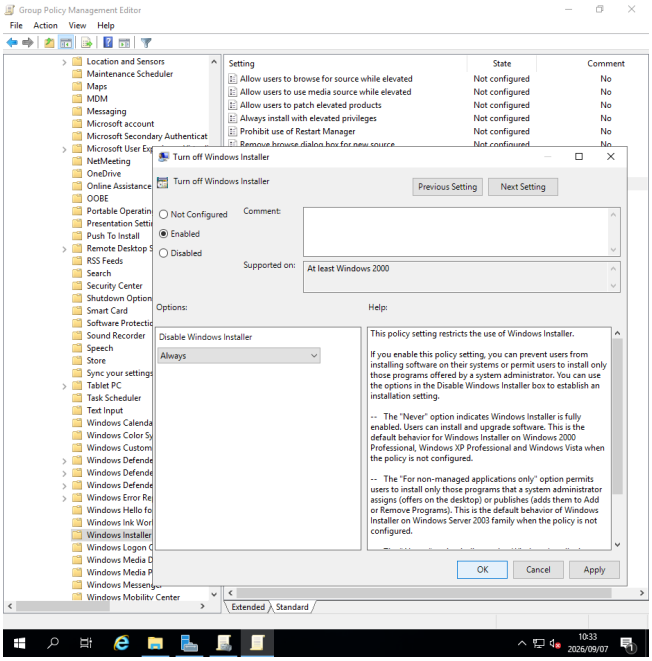


Figure 4.39: Turn off Windows Installer Enabled.

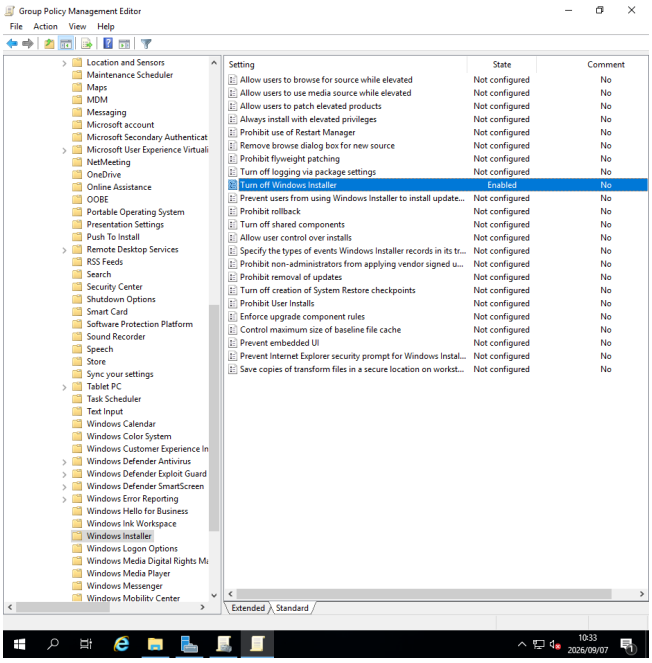


Figure 4.40: Turn off Windows Installer shown as enabled.

4.6 Testing

Testing is a critical step in the Group Policy configuration process to ensure that the policies are applied correctly and functioning as intended. After creating and linking the GPOs, administrators can use pre-deployment simulation tools to test the policies before applying them in a production environment. Additionally, administrators can

use the ‘gpresult‘ command to verify that the policies are being applied to the target users and computers. Furthermore, administrators can log in as a user affected by the GPOs to confirm that the restrictions and settings are enforced. The first and last methods of testing are quiet easily understood and therefore will not be covered in this report. The second method will be covered in the following section.

4.7 Chapter Summary

5. Conclusion

5.1 Summary

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5.2 Key Findings

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5.3 Future Work

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A. Appendix

A.1 Github Repository